

Approaches to Clinical Shock

Shock is the rude unhinging of the machinery of life.

Samuel Gross, MD ([a](#))

One of the recognizable features of the clinical condition known as "shock" is its name, which lacks any scientific or pathological merit, and is shared by nonclinical entities as far-removed as military operations ("shock and awe"). The clinical use of this term can be traced back to 1731, when a French surgeon named Henri LeDran published his experience with gunshot wounds, and used the term "chock" to describe the injuries that often had fatal outcomes ([1](#)). In a subsequent English translation of LeDran's account, "shock" was mistranslated as indicating the clinical course of the injuries rather than the injuries themselves. Thereafter, "clinical" shock became synonymous with a life-threatening and rapidly deteriorating condition. However, shock is a consequence of multiple pathological entities, and finding a common ground can be challenging.

This chapter is the first in a series of chapters devoted to clinical shock, and attempts to present a unified view of shock, including what it is, how it is classified, how it presents, and how it is managed. This is followed by three chapters devoted to the major shock syndromes: i.e., hypovolemic, cardiogenic, and septic shock. Simply stated, these conditions are the most challenging you will encounter in the ICU.

WHAT IS CLINICAL SHOCK?

The definition of clinical shock has had many iterations, including the colorful description in the introductory quote by Samuel Gross, one of the titans of surgery in the 19th century who included this description of shock in his popular textbook ([a](#)). An updated and improved definition of shock would state that *shock is a life-threatening circulatory disorder that is characterized by inadequate cellular oxygen utilization* ([2](#)). Shock can be the result of inadequate tissue oxygenation, or inadequate processing of oxygen by mitochondria, but the result is the same; i.e., there is a deficiency in oxidative ATP production. The clinical consequence of this derangement is dysfunction in one or more vital organs, which can progress to multiorgan failure and a fatal outcome.

Clinical Features

The clinical manifestations of shock typically include the following:

- . A decrease in blood pressure, which typically results in *hypotension*, defined as a systolic pressure <90 mm Hg or a mean pressure <70 mm Hg (2). (See the next section for more on hypotension in clinical shock.)
- . An increase in the plasma lactate level (>2 mmol/L), which is considered evidence of impaired cellular O₂ utilization (although other explanations are available, as described in Chapter 9.)
- . Clinical signs of organ dysfunction, which usually begins as *oliguria* (urine output <0.5 mL/kg per hour) and *altered mentation* (e.g., agitation, confusion, or depressed consciousness), and can progress to involve any and all major organs.
- . In cases of “low-flow shock”, the skin may be cool to touch, and there may be cyanotic changes in the distal extremities. Progression of the low-flow state can produce patchy cyanosis known as “mottled skin” or *livedo reticularis*.

Hypotension

Hypotension has been one of the standard clinical criteria for the diagnosis of clinical shock, but the reliability of hypotension as a marker of clinical shock is questionable. For example, clinical studies have shown that the blood pressure can be normal when there is evidence of inadequate tissue oxygenation (using lactate levels or the central venous O₂ saturation) (3). Furthermore (as anyone who has spent some time in an ICU can verify), hypotension is not always accompanied by evidence of inadequate O₂ utilization in tissues. As a result of observations like these, some consensus guidelines have recommended that hypotension should be abandoned as a marker of clinical shock (4).

Types of Shock

There are 4 different types of shock (shock syndromes), and these can be identified using the following hydraulic relationships: i.e., steady flow (Q) through a closed hydraulic circuit is directly related to the pressure gradient across the circuit (P_{in} – P_{out}), and is inversely related to the resistance to flow (R) through the circuit.

$$Q = (P_{in} - P_{out})/R \quad (14.1)$$

If the hydraulic circuit is the circulatory system, then flow is the cardiac output (CO), the inflow pressure is the mean arterial pressure (MAP), the outflow pressure is the right atrial pressure (RAP), and the resistance to flow is the systemic vascular resistance (SVR). Equation 14.1 is now expressed as follows:

$$CO = (MAP - RAP)/SVR \quad (14.2)$$

Since a decrease in blood pressure is commonplace in shock, the terms in the above equation can be rearranged to identify the determinants of mean arterial pressure (which is the driving pressure for blood flow):

$$\text{MAP} = (\text{CO} \times \text{SVR}) + \text{RAP} \quad (14.3)$$

The different shock syndromes are then identified by the determinants of MAP: i.e.,

- a. Low RAP = hypovolemic shock
- b. Low CO = cardiogenic shock, obstructive shock
- c. Low SVR = vasodilatory shock

The pattern of hemodynamic changes in each type of shock is shown in [Table 14.1](#), along with the reported prevalence of each (2). Note that 3 of the 4 shock syndromes are associated with a low cardiac output and a high SVR. The increase in SVR (i.e., vasoconstriction) is a compensatory response to the low-flow state, and limits the drop in blood pressure. This response is due to increased activity in the sympathetic nervous system, and activation of the renin-angiotensin-aldosterone system.

TABLE 14.1 Hemodynamic Patterns in the Shock Syndromes			
Shock Syndromes[†]	CVP	Cardiac Output	SVR
Hypovolemic (16%)	Low	Low	High
Obstructive (2%)	High	Low	High
Cardiogenic (16%)	High	Low	High
Vasodilatory (66%)	Low	Normal or High	Low

[†]Numbers in parentheses indicate relative frequency of occurrence (from Reference 2). CVP = central venous pressure, SVR = systemic vascular resistance.

Hypovolemic Shock

Hypovolemic shock is almost always the result of blood loss, and it usually appears when the decrease in blood volume exceeds 30% (5). Dehydration (e.g., from vomiting or diarrhea) does not typically result in shock because the fluid loss leads to an increase in the colloid osmotic pressure of plasma, which draws interstitial fluid into the vascular compartment. Decreases in blood pressure from dehydration are easily corrected with intravenous fluids.

Cardiogenic Shock

Shock from cardiac pump failure is most often the result of acute coronary syndromes, which account for almost half of the cases of cardiogenic shock (6). Less frequent causes include non-ischemic cardiomyopathies (28% of cases) and valvular dysfunction or arrhythmias (17% of cases). Cardiogenic shock presents the greatest challenge in shock management; i.e., despite advances in revascularization and mechanical cardiac support, the mortality rate remains at about 50% (7).

Obstructive Shock

Obstructive shock is caused by conditions that impede or block cardiac filling, such as massive pulmonary embolism, tension pneumothorax, and cardiac tamponade. This condition is

uncommon, and accounts for only 2% of cases of clinical shock (2). Treatment is directed at the responsible condition.

Vasodilatory Shock

Vasodilatory shock (also known as “distributive shock”) is a condition of widespread vasodilation involving both arteries and veins. It is the most frequently encountered type of shock (see Table 14.1), and septic shock accounts for most cases. *Septic shock is the leading cause of in-hospital deaths in the United States (8), and is also considered the leading cause of death worldwide (9).* Less common sources of vasodilatory shock include anaphylactic shock, spinal shock, and adrenal crisis.

BLOOD PRESSURE MONITORING

Blood pressure monitoring is mandatory in clinical shock, and direct intra-arterial recordings are preferred.

Direct Recordings

Direct arterial pressure recordings are obtained from the radial, brachial, axillary, or femoral arteries, and the site of cannulation can influence the recorded waveform (see next) .

Arterial Pressure Waveform

The arterial pressure waveform changes as it moves away from the proximal aorta, as shown in Figure 14.1. Note that as the pressure wave moves toward the periphery, the systolic pressure gradually increases and the systolic portion of the waveform narrows. The systolic pressure can increase as much as 20 mm Hg from the proximal aorta to the radial or femoral arteries (10). This increase in peak systolic pressure is offset by the narrowing of the systolic pressure wave, so the mean arterial pressure remains unchanged.

Systolic Amplification

The increase in systolic pressure in peripheral arteries is the result of pressure waves that are reflected back from vascular bifurcations and narrowed blood vessels (11). Reflected waves move faster when the arteries are stiff, and they reach the arterial pressure waveform before it has time to decrement. The convergence of antegrade and retrograde pressure waves then serves to heighten the peak of the pressure waveform. (You can see this effect when ocean waves meet from opposing direction. This magnification effect is implicated in the formation of “monster waves”.) Because systolic amplification is the result of retrograde pressure waves, it increases left ventricular afterload, but does not promote systemic blood flow.

Mean Arterial Pressure

The mean arterial pressure (MAP) is the preferred pressure for hemodynamic monitoring for two reasons: it is the principal driving force for systemic blood flow (10,12), and it does not change as the pressure waveform moves peripherally (unlike the systolic pressure). The MAP is measured electronically as the area under the arterial pressure wave, divided by the duration of the cardiac cycle. The management of clinical shock typically uses a target MAP of ≥ 65 mm Hg

(2).

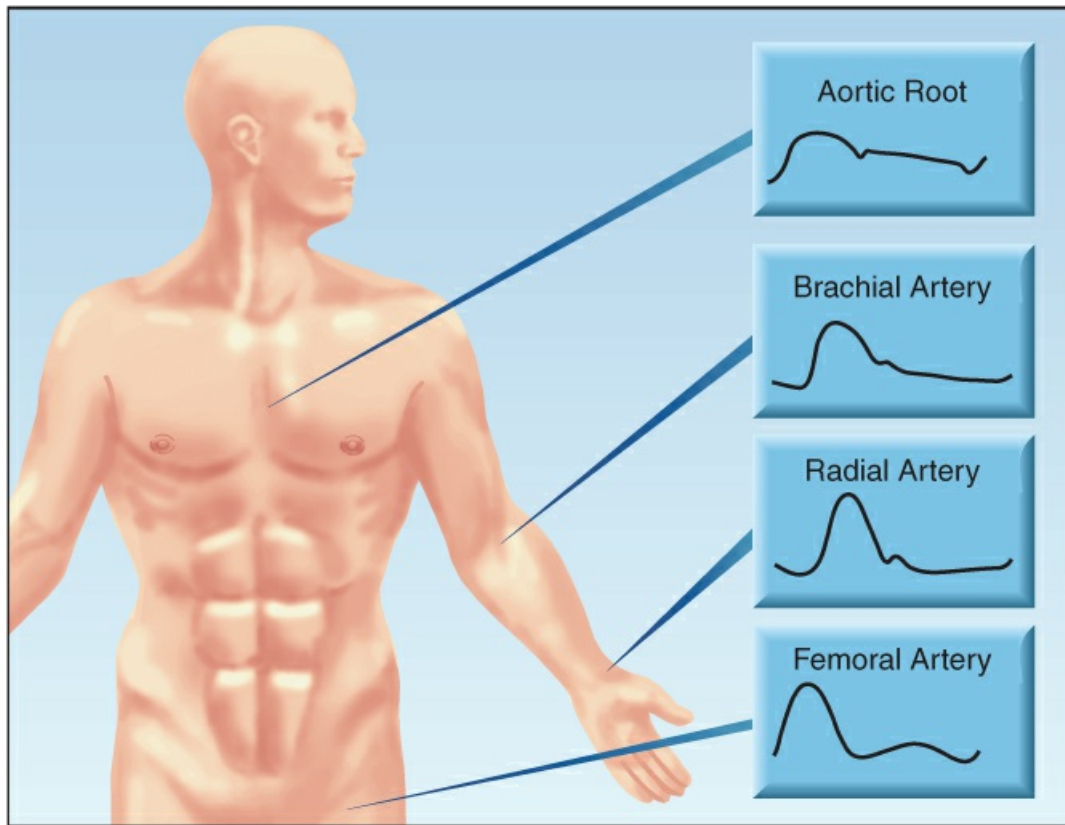


FIGURE 14.1 Arterial pressure waveforms at specific points in the arterial circulation.

Indirect Measurements

Direct blood pressure recordings are not always available, or feasible, and a brief description of indirect pressure measurements is therefore warranted.

The Principle

The indirect blood pressure (BP) measurement uses an inflatable bladder (attached to the underside of a cloth sleeve) that is wrapped around the upper arm or thigh in an area that overlies a major artery. The bladder in the sleeve is then inflated to compress the underlying artery. The effects of arterial compression are illustrated in [Figure 14.2](#). As the cuff pressure increases and the underlying artery is compressed, the pulsations in the artery gradually increase and then decrease until the artery is occluded. These “counterpulsations” produce oscillations in the bladder pressure, and the automated blood pressure devices (which are standard in ICUs) then measure these oscillations and convert them into measures of systolic, diastolic, and mean pressures. This is the *oscillometric method* of measuring the BP. The counterpulsations can also be converted into sound waves, which is the basis for the older *auscultation method* of measuring the BP.

Common Source of Error

Counterpulsations are more reproducible, and BP measurements are more reliable, when an

artery is compressed uniformly. Uniform compression is more likely if the size of the bladder is appropriate for the circumference of the upper arm. The optimal relationships between the height and width of the cuff bladder and the upper arm circumference are shown in [Figure 14.3](#). The following are some important points about the relationship between bladder size and arm size:

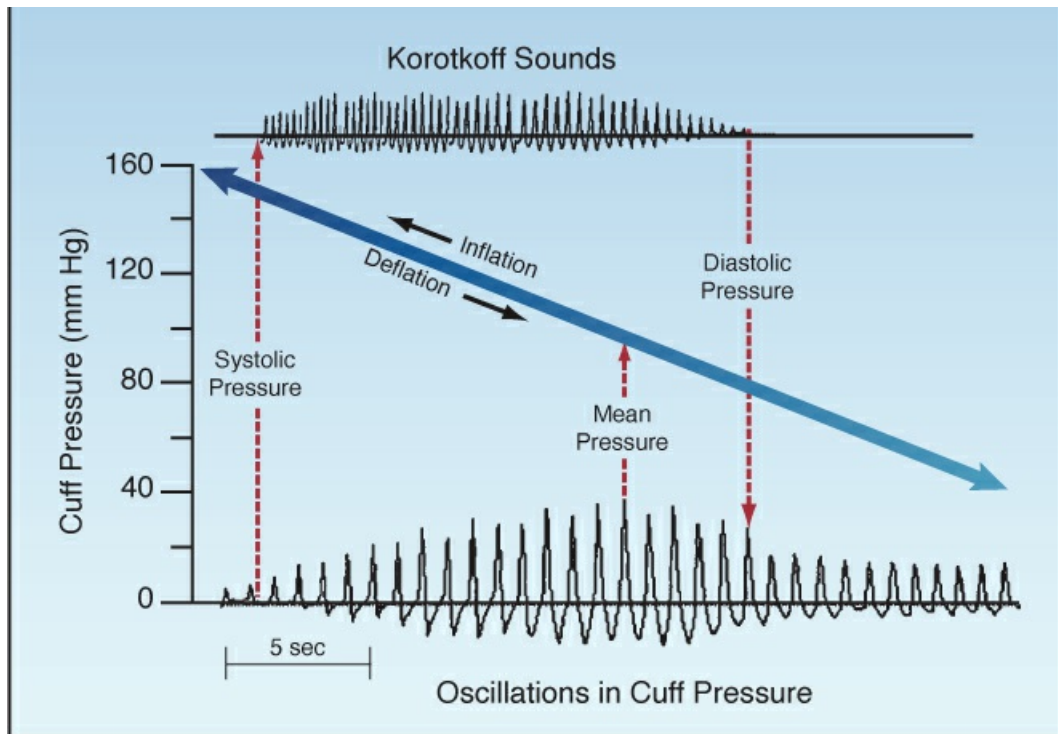


FIGURE 14.2 The oscillometric and auscultatory methods of measuring the blood pressure. See text for explanation.

- . To produce uniform arterial occlusion, the length of the inflatable bladder should be at least 80% of the circumference of the upper arm (measured midway between the shoulder and elbow), and the width of the bladder should be at least 40% of the upper arm circumference ([13](#)).
- . If the inflatable bladder is too small for the size of the upper arm, the pressure measurements will be falsely elevated ([1](#)). (Errors in measurement are much less pronounced when the bladder is too large relative to arm circumference.)
- . *The use of inappropriately-sized arm cuffs (and their inflatable bladders) is the most common source of error in indirect BP measurements ([13,14](#)).* In one study involving ICU patients, two-thirds of the indirect BP measurements were obtained with a cuff size that was inappropriate, and in these cases, 62% of the BP measurements differed from the direct, intra-arterial measurements by more than 10 mm Hg ([14](#)).

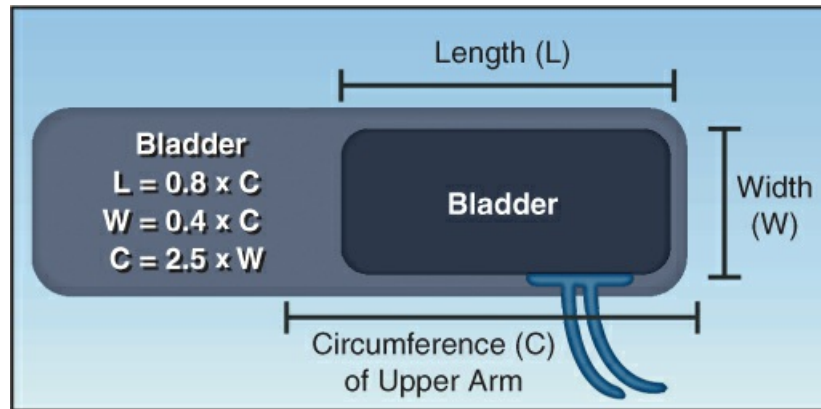


FIGURE 14.3 Optimal relationships between the height (H) and width (W) of the cuff bladder and the circumference (C) of the upper arm.

To help in selecting the appropriate cuff size, [Table 14.2](#) shows the recommended cuff sizes for different sizes of the upper arm (or thigh). An easier way to determine the appropriate cuff size is described next.

TABLE 14.2 Appropriate Size of Blood Pressure Cuff in Relation to Upper Arm Circumference		
Upper Arm Circumference	Blood Pressure Cuff	
	Size	Dimensions
22 to 26 cm	Small Adult	12 x 24 cm
27 to 34 cm	Adult	16 x 30 cm
35 to 44 cm	Large Adult	16 x 36 cm
45 to 52 cm	Adult Thigh	16 x 42 cm

From Reference 13.

Simple Method for Assessing Cuff Size

Align the cuff so that the long axis runs along the long axis of the arm. Then turn the cuff over so the bladder on the underside of the cuff is facing upward, and wrap the cuff around the upper arm. The bladder (width) should encircle close to half (40%) of the upper arm. If the bladder encircles less than half of the upper arm, the cuff is too small, and a larger cuff should be used. No change in cuff size is needed if the bladder is too big (i.e., if it encircles most of the upper arm), because this discrepancy does not produce significant errors in measurement ([13](#)).

Accuracy

The limited accuracy of indirect BP measurements in critically ill patients is well documented ([14,15](#)). This is demonstrated in a recent meta-analysis of 7 studies comparing automated and direct BP measurements in ICU patients ([15](#)), which showed that the automated BP measurement could differ from the direct BP by as much as 55 mm Hg (from –15 mm Hg to +40 mm Hg) in individual patients. This is an unacceptable range of variation, and it demonstrates the need for

direct BP recordings in the critically ill.

TISSUE PERFUSION AND OXYGENATION

The approach to shock requires more than blood pressure monitoring, as measures of tissue perfusion and oxygenation are essential for both the detection and management of shock. These measures are shown in [Table 14.3](#), along with the threshold value of each that indicates possible or probable shock. Each of these measures has been described elsewhere in the book, and the following represents only a brief summary.

Oxygen Transport Parameters

There are three measures of systemic oxygen transport: oxygen delivery (DO_2), oxygen uptake (VO_2), and oxygen extraction (VO_2/DO_2). These measures, which are described in detail in [Chapter 9](#), provide an assessment of the global balance between O_2 supply and O_2 demand, which is deranged in all forms of shock.

Cardiac Index

Oxygen transport is dependent on the cardiac output, which is most accurately measured by the thermodilution technique using pulmonary artery catheters (see [Chapter 8](#)). Alternative methods include the stroke volume determination from arterial pressure waveforms (see [Figure 11.5](#)) or measurements of aortic flow velocity using Doppler ultrasound. (These are described in [Chapter 11](#).) The stroke volume is multiplied by the heart rate to obtain the cardiac output.

As explained in [Chapter 8](#), hemodynamic measurements are typically expressed in relation to body size using the body surface area in meters squared. The size-adjusted cardiac output is called the *cardiac index* (CI), and the normal range is 2.4–4 L/min/m² (see [Table 8.1](#)). A subnormal CI (<2.5 L/min/m²) is considered evidence of inadequate tissue perfusion.

Oxygen Delivery

The rate of oxygen delivery in arterial blood (DO_2) is derived as the product of the cardiac index (CI) and the oxygen concentration in arterial blood (CaO_2); i.e.,

$$\text{DO}_2 = \text{CI} \times \text{CaO}_2 \quad (14.4)$$

(For a more detailed expression of this equation, see [Equation 9.8](#).) *A decrease in O_2 delivery is the culprit in three of the four types of clinical shock (i.e., hypovolemic, cardiogenic, and obstructive shock).* The DO_2 is normally 520–600 mL/min/m² (see [Table 9.2](#)), and a DO_2 of 300 mL/min/m² has been identified as the threshold for inadequate tissue oxygenation ([16](#)). However this is not a consistent finding, and some studies have not been able to identify a DO_2 that impairs aerobic metabolism ([17](#)).

Oxygen Uptake

The rate of oxygen uptake into tissues (VO_2), which is equivalent to the O_2 consumption, is calculated as the product of the cardiac index and the difference in O_2 concentration between

arterial and venous blood ($\text{CaO}_2 - \text{CvO}_2$); i.e.,

$$\text{VO}_2 = \text{CO} \times (\text{CaO}_2 - \text{CvO}_2) \quad (14.5)$$

(For a more detailed expression of this equation, see [Equation 9.10](#).) The VO_2 is normally 110–160 mL/min/m², and an abnormally low VO_2 (<110 mL/min/m²) is evidence of impaired O_2 utilization (in the absence of aerobic hypometabolism, which is uncommon in critically ill patients).

Oxygen Extraction

Oxygen extraction is the ratio of O_2 uptake to O_2 delivery, which identifies the fraction of delivered oxygen that is taken up into the tissues.

$$\text{O}_2 \text{ Extraction} = \text{VO}_2 / \text{DO}_2 \quad (14.6)$$

When the arterial O_2 saturation is close to 100%, [Equation 14.6](#) can be reduced to variables that are easily monitored; i.e.,

$$\text{O}_2 \text{ Extraction} = (\text{SaO}_2 - \text{ScvO}_2) \quad (14.7)$$

where SaO_2 and ScvO_2 are the arterial and central venous O_2 saturations, respectively. (For the derivation of this equation, see Equations 9.12–9.14.) This expresses the O_2 extraction as the extent of hemoglobin desaturation as blood flows through the capillaries.

The O_2 extraction is normally 0.2–0.3 (20–30%), and it can increase to about 0.5 (50%) in response to a decrease in DO_2 , which helps to keep the VO_2 constant (see [Figure 9.3](#)). Further decreases in DO_2 beyond this point will not elicit the compensatory increase in O_2 extraction, and the VO_2 will begin to fall (indicating the onset of anaerobic metabolism). Therefore, an O_2 extraction of $\geq 50\%$ indicates that O_2 delivery is low enough to threaten, or impair, aerobic metabolism (18). There are also conditions that impair O_2 extraction (e.g., sepsis, liver failure), and these conditions can be identified by an O_2 extraction that is fixed (i.e., does not increase in response to a decrease in DO_2), or is $\leq 20\%$. In this situation, the O_2 extraction is not a reliable indicator of changes in O_2 delivery.

Central Venous O_2 Saturation

When the SaO_2 is constant, or is close to 1.0 (100%), [Equation 14.7](#) can be rewritten as:

$$\text{O}_2 \text{ Extraction} = (1 - \text{ScvO}_2) \quad (14.8)$$

Thus, the central venous O_2 saturation (ScvO_2) can be used alone to monitor O_2 extraction (19). The normal ScvO_2 is about 65–75%, and an increase in O_2 extraction will drop the ScvO_2 below this range. The maximal O_2 extraction (about 50%) corresponds to an ScvO_2 of about 50%, and thus an ScvO_2 of $\leq 50\%$ can be used as a marker of threatened or impaired tissue oxygenation (20).

At the other end of the spectrum, an ScvO₂ that is abnormally high ($\geq 80\%$) is evidence of impaired tissue O₂ extraction, which is a characteristic feature of septic shock. In this situation, the ScvO₂ can also be fixed (i.e., does not vary with changes in O₂ delivery), and if this occurs, the ScvO₂ is not a reliable indicator of changes in O₂ delivery. Enter the PCO₂ gap (see next).

TABLE 14.3 Global Measures of Tissue Perfusion and Oxygenation		
Measure	Critical Range	Interpretation
Cardiac Index (CI)	<2.5 L/min/m ²	Inadequate Perfusion
<i>Oxygen Transport</i>		
Oxygen Delivery (DO ₂)	≤ 300 mL/min/m ²	Inadequate O ₂ Delivery
Oxygen Uptake (VO ₂)	≤ 110 mL/min/m ²	Inadequate O ₂ Utilization [†]
Oxygen Extraction (VO ₂ /DO ₂)	$\geq 50\%$ $<20\%$	Inadequate O ₂ Delivery Inadequate O ₂ Extraction
Central Venous O ₂ Saturation (ScvO ₂)	$<50\%$ $>80\%$	Inadequate O ₂ Delivery Inadequate O ₂ Extraction
PCO ₂ Gap	>6 mm Hg (>0.8 kPa)	Inadequate Perfusion
Plasma Lactate	>2 mmol/L	Inadequate O ₂ Utilization [†]

[†]Present in all forms of shock.

The PCO₂ Gap

The venoarterial PCO₂ difference, also known as the PCO₂ gap, is an indirect measure of CO₂ washout from tissues (i.e., tissue perfusion). The PCO₂ gap is normally 2–5 mm Hg (0.3–0.7 kPa), and an increase in the gap to >6 mm Hg (>0.8 kPa) is evidence of tissue hypoperfusion (21). Unlike the ScvO₂, *the PCO₂ gap is dependent only on tissue perfusion, and it provides no information about systemic oxygenation.* The PCO₂ gap is easily obtained, and should be monitored in all patients with shock, especially those with low-flow shock (hypovolemic, cardiogenic, and obstructive shock). It can also be useful for detecting a low-flow state in septic shock, which is usually seen in advanced cases. (*Caveat:* A metabolic acidosis can increase the PCO₂ gap. This is the result of an increase in nonmetabolic CO₂ production, and it could be misinterpreted as evidence of tissue hypoperfusion. When a metabolic acidosis is present, only a normal PCO₂ gap has relevance.)

Plasma Lactate

The plasma lactate level is considered one of the most important measures in clinical shock, and has both diagnostic and prognostic applications. As mentioned earlier, a plasma lactate >2 mmol/L is considered evidence of inadequate cellular O₂ utilization, in the absence of conditions known to cause “aerobic hyperlactatemia” (see Table 9.5). If shock is the cause of an elevated

lactate level, then failure to normalize the level within 24 hours has negative implications for a satisfactory outcome (see [Figure 9.4](#)). (The plasma lactate level is described in detail in [Chapter 9](#).)

GENERAL MANAGEMENT

The management of clinical shock has two components: treating the condition responsible for the shock, and generalized hemodynamic management. The latter component is the focus here.

Volume Resuscitation

The infusion of intravenous fluids is the very first step in the management of shock, regardless of the type of shock. (This may seem counterproductive in cardiogenic shock, but cardiac filling pressures are often suboptimal in acute heart failure, especially when it involves the right side of the heart.) Crystalloid fluids are preferred, even though about 75% of infused crystalloid fluids expand the interstitial volume, not the plasma volume (see [Chapter 11](#)).

Infused Volume

The initial infusion volume can depend on several variables (e.g., type of shock, type of patient). For the most common shock syndrome (septic shock), the recommended resuscitation volume is ≥ 30 mL/kg, using *ideal body weight*, in the first 3 hours ([22](#)). The minimum infusion volume for a 70 kg adult would then be about 2 liters, and for a crystalloid fluid, this means about 500 mL (25%) will be added to the plasma volume, and the remaining 1.5 liters (75%) will contribute to edema formation. This increment in plasma volume represents about 15% of the plasma volume for a 70 kg adult male (whose plasma volume is about 40 mL/kg), which seems reasonable if you consider that a 15% drop in blood volume is the threshold for producing symptoms (see next chapter). However, this “one size fits all” approach may not be suitable for adults of varying body size, and for females (whose plasma volume is lower than in males). The following are some recommendations for the initial fluid resuscitation:

- . The initial resuscitation volume should take into account the type of shock (if known) and the type of patient (e.g., body size and gender), and should probably should not exceed 2 liters (crystalloid fluid) over 1–2 hours.
- . The immediate goal is restoration of the baseline blood pressure.
- . If the blood pressure goal is not achieved and there is other evidence of circulatory compromise (e.g., low urine output, mental status change, or elevated lactate level), then infusion of a vasoconstrictor agent (vasopressor) should be initiated.
- . Volume resuscitation beyond the initial period should be guided by the assessment of fluid responsiveness (described in [Chapter 11](#)).

Vasopressor Infusions

The first issue that surfaces for vasopressor infusions is the traditional recommendation that vasopressor agents should be infused through a central venous catheter (to minimize the risk of drug extravasation and tissue necrosis). This is a source of angst when the vasopressor infusion is

an immediate need and there is no indwelling central line. Fortunately, at least 16 clinical studies have shown that vasopressors can be infused safely through peripheral catheters (in the upper arm) for up to 48 hours (23), and vasopressor infusions through midline catheters have continued for 7 days without incident (24). If there is a problem securing a free-flowing peripheral catheter, then insertion of an intraosseous catheter will provide a temporary fix (see [Chapter 1](#)).

Drug Extravasation

Extravasation of a vasoconstrictor drug should be managed as follows:

- . When extravasation is first evident, stop the infusion immediately, disconnect the IV line and attach it to a syringe and gently aspirate the remaining solution in the catheter and whatever extravasated fluid can be retrieved. Then remove the catheter. Do not flush the line prior to removal.
- . The drug that is used in this situation is phentolamine, a long-acting alpha blocker. The dose is 5–10 mg in 10 mL isotonic saline, which is injected directly into the area of extravasation (25). If there is skin blanching (from the local vasoconstriction), it should disappear almost immediately.
- . Phentolamine must be used within 12 hours of the event to be successful.

Target Blood Pressure

The consensus recommendation is that vasopressor infusions should be titrated to maintain a mean arterial pressure (MAP) of at least 65 mm Hg (2,4,22), which is the minimum pressure needed to maintain cerebral autoregulation (the process that maintains cerebral blood flow in the face of declining blood pressures) (26). The concern that a higher target pressure might be necessary in patients with longstanding hypertension has not been validated in clinical studies (27).

VASOPRESSOR AGENTS

A variety of vasopressors are available, and are used primarily in vasodilatory (septic) shock. Most are catecholamines that promote vasoconstriction by stimulating α_1 -type adrenergic receptors. The vasopressors that are currently available are listed in [Table 14.4](#), along with the dose range for each agent. These agents are given by continuous infusion (without a loading dose), starting at a low dose and titrating upward as needed. Resistance to high doses of a vasopressor usually prompts the addition of a second agent, although there is no evidence of a survival benefit from this practice (see A FINAL WORD at the end of the chapter).

Norepinephrine

Norepinephrine is the most widely used vasopressor, and is the preferred vasopressor in septic shock (22,28). Its principal action is α_1 -receptor-mediated vasoconstriction, but it is also a weak β -receptor agonist, which produces some degree of cardiac stimulation. This might explain why the vasoconstriction from norepinephrine (in the usual doses) does not impair renal function (29).

Dosing Regimen

Norepinephrine infusions are usually started at a rate of 5–10 µg/min, which is titrated upward as needed. The β-receptor stimulation can increase the cardiac output in the low-dose range (<10 µg/min), while vasoconstriction from α-receptor stimulation begins to predominate at dose rates above 10 µg/min. When the dose reaches 30 µg/min, further increases are unlikely to produce more of a vasoconstrictor response, and a second vasopressor can be added at this point.

Adverse Effects

Serious adverse events related to norepinephrine infusions include ischemic events (myocardial, cerebral, or limb ischemia) and troublesome arrhythmias (tachycardias and bradycardia). These have been reported in about 10% of patients (30), but it is possible that many of these complications are the result of the shock, and not the vasopressor.

IMMUNOSUPPRESSION: There is evidence that leukocyte activation by bacterial products (which is an important part of the immune response) is suppressed during norepinephrine infusions (31). This is attributed to the binding of norepinephrine to adrenergic receptors on the surface of leukocytes, and it raises the possibility that norepinephrine acts as an immunosuppressant, and increases the risk of infection. The clinical significance of this is unclear, but the ironic significance is quite clear, considering the popularity of norepinephrine for treating septic shock.

TABLE 14.4

Parenteral Vasopressor Agents

Agent	Dose	Comments
Norepinephrine	5–40 µg/min	The most widely used vasopressor in clinical shock
Epinephrine	0.1–0.5 µg/kg/min	Drug of choice for anaphylactic shock, and second-line agent for septic shock Promotes lactate production
Dopamine	5–50 µg/kg/min	Once popular, but hampered by undesirable cardiac stimulation
Phenylephrine	0.5–6 µg/kg/min	A pure α agonist used mostly for anesthesia-related hypotension
Vasopressin	0.01–0.04 Units/hr	Used as a second agent in cases of septic shock that are resistant to norepinephrine
Angiotensin II	0.02–0.08 µg/kg/min initially, then infuse at ≤0.04 µg/kg/min	Can promote venous thrombosis

Epinephrine

Epinephrine is a potent β-receptor agonist that produces dose-dependent increases in heart rate, stroke volume, and blood pressure (32). It has an important role in cardiopulmonary resuscitation (see Chapter 21), and in the resuscitation of anaphylactic shock (see Chapter 17), but it is a second-line vasopressor in septic shock, and is reserved for cases where hypotension is refractory

to norepinephrine.

Dosing Regimen

The recommended doses for epinephrine in cardiopulmonary resuscitation and anaphylactic shock are presented in Chapters 21 and 17, respectively. For septic shock, epinephrine is given as a continuous infusion with no loading dose, starting at 0.1 µg/kg/min and titrating upward, as needed. The maximum dose is 0.5 µg/kg/min (27).

Adverse Effects

The adverse effects of epinephrine include unwanted cardiac stimulation (i.e., tachycardia and arrhythmias), and splanchnic hypoperfusion from α-mediated vasoconstriction (33), which can damage the mucosal barrier in the bowel. Epinephrine also has undesirable metabolic effects, including an increase in metabolic rate (which is counterproductive in a shock syndrome), hyperglycemia (from α-receptor-mediated inhibition of insulin secretion), and an increase in plasma lactate levels (due to an increase in aerobic lactate production) (34). The increase in lactate levels can be misleading in shock syndromes.

Dopamine

Dopamine is an endogenous catecholamine that serves as a precursor for norepinephrine. It was once the preferred vasopressor in septic shock, but its tendency to produce troublesome tachycardia and arrhythmias led to a fall from grace. Dopamine has a variety of dose-dependent effects (35):

- . At low infusion rates (≤ 3 µg/kg/min), dopamine selectively activates dopamine-specific receptors in the renal and splanchnic circulations, resulting in increased blood flow in these regions.
- . At moderate infusion rates (3–10 µg/kg/min), dopamine stimulates β-receptors in the heart and peripheral circulation, producing an increase in cardiac stroke output.
- . At high infusion rates (>10 µg/kg/min), dopamine produces a dose-dependent activation of α-receptors in the peripheral circulation, resulting in progressive vasoconstriction and an increase in systemic vascular resistance. This vasopressor effect increases ventricular afterload, and can reverse the stroke volume augmentation produced by lower doses of dopamine.

Dosing Regimen

Dopamine is usually started at a dose of 5 µg/kg/min, and is titrated upward as needed. A dose of 5–10 µg/kg/min will usually increase the cardiac output, but a higher dose (>10 µg/kg/min) may be needed to increase the blood pressure. The benefits of dopamine are usually seen at dose rates of 5–20 µg/kg/min, but dose rates of 50 µg/kg/min and even higher have been reported (35).

Adverse Effects

The major risk with dopamine is tachycardia and troublesome tachyarrhythmias (e.g., atrial fibrillation), which occur more frequently with dopamine than with the other vasopressors (36). As mentioned, this risk has resulted in a marked decline in the popularity of dopamine for both cardiogenic and septic shock. Another risk with dopamine that deserves mention is increased

intraocular pressure, which has been reported at low dose rates (1–4 µg/kg/min) (37).

Phenylephrine

Phenylephrine is a pure α -receptor agonist that produces widespread vasoconstriction, which can be accompanied by bradycardia, a decrease in cardiac stroke output (usually in patients with cardiac dysfunction), and hypoperfusion of the kidneys and bowel. The principal use of phenylephrine is for hypotension produced by anesthesia (spinal or general), where it is often given in bolus doses. Phenylephrine is not recommended for any of the shock syndromes (22,38), although a clinical study comparing phenylephrine and norepinephrine in septic shock showed no difference in hemodynamic effects or clinical outcomes with either drug (39). In the ICU, phenylephrine is useful for managing septic shock in patients with rapid atrial fibrillation (author's observation).

Dosing Regimens

- . For treating hypotension during anesthesia, phenylephrine can be given as a bolus dose of 50–100 µg, which can be repeated once after a few minutes, if needed. If the hypotension is not relieved, then a continuous infusion of the drug is recommended, starting at a dose rate of 10–35 µg/min and titrating upward. The maximum recommended dose is 200 µg/min (38).
- . For managing septic shock, phenylephrine is given as a continuous infusion with no initial bolus dose, starting at a dose of 0.5 µg/kg/min, and titrating upwards as needed. Dose rates above 6 µg/kg/min are unlikely to produce any further benefit (38).

Vasopressin

Vasopressin, also known as antidiuretic hormone, produces vasoconstriction as a result of specialized vasopressin (V_1) receptors on vascular smooth muscle. Vasoconstriction is most prominent in skin, skeletal muscle, and splanchnic circulations (39). Vasopressin does not increase blood pressure in healthy volunteers, but it can produce significant increases in blood pressure in patients with vasodilatory shock (40). Other actions of vasopressin include enhanced water reabsorption in the distal renal tubules (mediated by V_2 receptors), and stimulation of ACTH release by the anterior pituitary gland (mediated by V_3 receptors).

Use as a Vasopressor

Vasopressin can be used in the following clinical situations.

- . In the resuscitation of cardiac arrest, vasopressin can be given as a single IV dose (40 units) to replace the first or second dose of epinephrine.
- . In cases of septic shock that are resistant to hemodynamic support with norepinephrine (usually defined as norepinephrine dose of ≥ 0.2 µg/kg/min without an adequate blood pressure), the addition of vasopressin is recommended (22), as this can raise the blood pressure and reduce the catecholamine requirement (catecholamine sparing effect). However, the addition of vasopressin has *no proven survival benefit* (41).

Dosing Regimen

For septic shock, vasopressin is given as a continuous infusion at 0.03–0.04 Units/hr. This dose rate is not titrated (unlike the catecholamine vasopressors). The adverse effects of vasopressin are similar in type and frequency to those mentioned previously for norepinephrine (30).

Angiotensin II

Angiotensin II is a synthetic analogue of human angiotensin that acts on vascular smooth muscle to promote systemic vasoconstriction, and it also stimulates vasopressin release from the posterior pituitary gland. It was approved for use in vasodilatory shock in 2018, and has been used primarily in cases of norepinephrine resistance. It has raised the blood pressure in as many as 67% of cases (42), but the impact on mortality rates has been inconsistent (42,43).

Dosing Regimen

Angiotensin II is given by continuous intravenous infusion: the initial dose is 20 ng/kg/min (0.02 µg/kg/min), which can be increased every 10–15 minutes if needed, to 80 ng/kg/min (0.08 µg/kg/min) in the first 3 hours (44). Thereafter, the maintenance dose should not exceed 40 ng/kg/min (0.04 µg/kg/min). There is no dose adjustment for hepatic or renal dysfunction.

Adverse Effects

The major risk with angiotensin II is venous thrombosis. Angiotensin stimulates the release of plasminogen activator inhibitor-1 from vascular smooth muscle, which blocks fibrinolysis, and it also increases platelet adhesion. In one clinical trial, the incidence of thrombotic events with angiotensin II was more than double the incidence with placebo (13% vs. 5%, respectively) (44).

Midodrine

Midodrine is an orally administered α -receptor agonist that was FDA-approved in 1996 for the treatment of orthostatic hypotension. The clinical experience with midodrine in the ICU is summarized below.

- . Midodrine has proven effective for limiting or preventing hypotension during hemodialysis (45). The effective dose varies from 5 mg to 20 mg, given 20–30 minutes before dialysis. An additional smaller dose can be given during the dialysis if needed.
- . Adjuvant treatment with midodrine (20 mg every 8 hours) does not hasten the discontinuation of parenteral vasopressor therapy (46,47).

Despite its appeal as a vasopressor that is easy to administer, midodrine offers no advantage for the management of shock.

GOALS OF RESUSCITATION

The goals of resuscitation for clinical shock have been mentioned throughout this chapter, and are briefly summarized here. The goals can be grouped into two categories, tissue perfusion and oxygen transport, as shown in Table 14.5.

Tissue Perfusion

The importance of blood flow in shock is readily apparent when you consider that three of the four types of shock (i.e., hypovolemic, cardiogenic, and obstructive) are low-flow conditions. One of the very first goals in shock resuscitation is to obtain accurate measurements of the mean arterial pressure (MAP), which is the pressure that drives peripheral flow. This requires direct intra-arterial pressure recordings from an arterial catheter. Once this is achieved the MAP should be maintained at ≥ 65 mm Hg. Do not monitor the systolic pressure (if possible), since this is a reflected pressure wave that is traveling back to the heart (i.e., away from the periphery). Remember that flow in the microcirculation is laminar, not pulsatile, and has no systolic and diastolic components.

Monitoring the cardiac output with pulmonary artery (PA) catheters was once a staple of critical care management, but this practice took a nosedive about 25 years ago because of evidence that PA catheters do not improve survival in critically ill patients. (A harsh judgement, since the catheter is a monitoring device, not a therapy.) If a cardiac output measurement is available, the goal is to keep the size-adjusted output (the cardiac index) above 2.5 L/min/m^2 .

An assessment of tissue perfusion is also possible using the PCO_2 gap, which is a measure of CO_2 washout from tissues. A PCO_2 gap that is <6 mm Hg is considered evidence of adequate tissue perfusion. This measure is easily obtained, and should be monitored in all patients with shock, even when the cardiac output measurement is available (since microcirculatory flow can be sluggish when the global cardiac output is in the normal range). *Caveat:* Metabolic acidosis can increase the nonmetabolic production of CO_2 , and this can result in a spurious increase in the PCO_2 gap.

Finally, tissue perfusion can be assessed by monitoring organ function, and urine output is appropriate for this purpose because a decrease in urine output is often one of the earliest signs of impending shock. Since oliguria is identified by a urine output $<0.5 \text{ mL/kg/hr}$, the goal of resuscitation is to exceed this threshold. Mental status can also be monitored, but it is not as easy to quantify as urine output.

TABLE 14.5 Goals of Resuscitation for Shock	
Category	Goals
Tissue Perfusion	MAP ≥ 65 mm Hg CI $\geq 2.5 \text{ L/min/m}^2$ PCO_2 Gap <6 mm Hg Urine Output $>0.5 \text{ mL/kg/hr}$
Oxygen Transport	$\text{DO}_2 >300 \text{ mL/min/m}^2$. $\text{VO}_2 >110 \text{ mL/min/m}^2$ O_2 Extraction $<50\%$ $\text{ScvO}_2 >50\%$ Lactate $<2 \text{ mmol/L}$

Oxygen Transport

Monitoring the O₂ transport variables has obvious implications because a derangement on one of these variables is responsible for clinical shock.

- . Three of the major shock syndromes (i.e., hypovolemic, cardiogenic, and obstructive shock) are the result of inadequate O₂ delivery to tissues, usually from an inadequate cardiac output, with a contribution from a low hemoglobin level in one case (hemorrhagic shock). The goal in these cases is a cardiac index (CI) >2.5 L/min/m², an O₂ delivery (DO₂) >300 mL/min/m², and an O₂ uptake (VO₂) >110 mL/min/m². If these variables are not monitored (which happens in most cases), then the goal of resuscitation is a central venous O₂ saturation (ScvO₂) >50% and a plasma lactate level <2 mmol/L.
- . *Septic shock is NOT the result of inadequate O₂ delivery, but instead is the result of a defect in oxidative phosphorylation in dysfunctional mitochondria (48,49).* This condition is known as *cytopathic hypoxia* (48), and it can be associated with a defect in O₂ extraction from capillary blood. The goals of management in this case include an O₂ extraction <80%, and an O₂ uptake (VO₂) >110 mL/min/m². (The ScvO₂ may not be reliable in sepsis, as a result of the defective O₂ extraction.) If these variables are not monitored, the goal of resuscitation is simply a normal plasma lactate level. (The management of septic shock is presented in detail in [Chapter 17](#).)
- . The universal goal of resuscitation in all forms of shock is normalization of the plasma lactate level (<2 mmol/L).

Of course, the tacit goal of resuscitation in clinical shock is survival.

A FINAL WORD

Hypotension as a Consequence, not a Cause, of Shock

The resuscitation of shock is dominated by the desire to raise the blood pressure, which is based on the assumption that the low blood pressure is responsible for the shock. The following observations create some doubt about this approach.

- . Shock is defined as a derangement in cellular O₂ utilization (2), not a derangement in blood pressure.
- . Relieving hypotension has been the major focus of shock management for over 70 years (since norepinephrine was introduced for clinical use in 1950), yet the mortality rate in shock remains unacceptably high, and the use of two and even three vasopressors offers no survival benefit (50).
- . Low blood pressures occur in conditions other than shock (e.g., autonomic neuropathies, adrenal insufficiency, 5% of healthy adults), and they do produce anoxic organ injury or elevated lactate levels. In fact, low blood pressures have proven beneficial in penetrating injuries by limiting blood loss (see the next chapter).

One possible explanation for these observations is that *hypotension does not cause shock, but instead is a consequence of shock*. Shock is well known for causing multiorgan dysfunction, and

one of the dysfunctional organs could be the blood vessels. If this is the case, then correcting the blood pressure will not correct the shock (or improve survival), just like hemodialysis for shock-related renal failure will not correct the shock (or improve survival).

References

- a. Gross SD. System of surgery: pathological, diagnostic, therapeutique, and operative, vol 1. 2nd ed., Philadelphia: Blanchard and Lea; 1862.
1. Manji R, Wood KE, Kuman A. The history and evolution of circulatory shock. *Crit Care Clin* 2009; 25:1–29.

What is Shock?

2. Vincent J-L, De Backer D. Circulatory shock. *N Engl J Med* 2013; 369:1726–1734.
3. Rivers E, Nguyen B, Havstad S, et al. Early goal-directed therapy in the treatment of severe sepsis and septic shock. *N Engl J Med* 2001; 345:1368–1377.
4. Cecconi M, De Backer D, Antonelli M, et al. Consensus on circulatory shock and hemodynamic monitoring. Task force of the European Society of Intensive Care Medicine. *Intensive Care Med* 2014; 40:1795–1815.
5. Advanced Trauma Life Support® Student Course Manual. 10th ed., Chicago: American College of Surgeons, 2012; 49–50.
6. Berg DD, Bohula EA, Morrow DA. Epidemiology and causes of cardiogenic shock. *Curr Opin Crit Care* 2021; 27:401–408.
7. Rob D, Bělohávek J. The mechanical support of cardiogenic shock. *Curr Opin Crit Care* 2021; 27:440–446.
8. Liu V, Escobar GJ, Greene JD, et al. Hospital deaths in patients with sepsis from 2 independent cohorts. *JAMA* 2014; 312:90–92.
9. Rudd KE, Johnson SC, Agesa KM, et al. Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study. *Lancet* 2020; 395:200–211.

Blood Pressure Monitoring

10. Augusto J-L, Teboul J-L, Radermacher P, Asfar P. Interpretation of blood pressure signal: physiological bases, clinical relevance, and objectives during shock states. *Intensive Care Med* 2011; 37:411–419.
11. Nichols WW, O'Rourke MF. McDonald's blood flow in arteries. 3rd ed. Philadelphia: Lea & Febiger, 1990; 251–269.
12. Shapiro DS, Loiacono LA. Mean arterial pressure: therapeutic goals and pharmacologic support. *Crit Care Clin* 2010; 26:285–293.
13. Pickering TG, Hall JE, Appel LJ, et al. Recommendations for blood pressure measurement in humans and experimental animals: Part 1: Blood pressure measurement in humans: a statement for professionals from the Subcommittee of Professional and Public Education of the American Heart Association Council on High Blood Pressure Research. *Circulation* 2005; 111:697–716.
14. Bur A, Hirschl M, Herkner H, et al. Accuracy of oscillometric blood pressure measurement according to the relation between cuff size and upper-arm circumference in critically ill patients. *Crit Care Med* 2000; 28:371–376.
15. Nedel WL, Vasconcellos AT, Gunsch KA, Soares PHR. Accuracy and precision of oscillometric noninvasive blood pressure measurement in critically ill patients: systematic review and meta-analysis. *Anaesthesiol Intensive Ther* 2022; 54:425–431.

Tissue Perfusion and Oxygenation

16. Komatsu T, Shibutani K, Okamoto K, et al. Critical level of O₂ delivery after cardiopulmonary bypass. *Crit Care Med* 1987; 15:194–197.
17. Mohsenifar Z, Goldbach P, Tashkin DP, et al. Relationship between O₂ delivery and O₂ consumption in the adult respiratory distress syndrome. *Chest* 1983; 84:267–272.
18. Wilkerson DK, Rosen AL, Gould SA, et al. Oxygen extraction ratio: a valid indicator of myocardial metabolism in anemia. *J Surg Res* 1987; 42:629–634.
19. Maddirala S, Khan A. Optimizing hemodynamic support in septic shock using central venous and mixed venous oxygen saturation. *Crit Care Clin* 2010; 26:323–333.
20. Vallet B, Robin E, Lebuffe G. Venous oxygen saturation as a physiologic transfusion trigger. *Crit Care* 2010; 14:213–217.
21. Scheeren TWL, Wicke JN, Teboul J-L. Understanding the carbon dioxide gaps. *Curr Opin Crit Care* 2018; 24:181–189.

General Management

22. Evans L, Rhodes A, Alhazzani W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock 2021. *Intensive Care Med* 2021; 47:1181–1247.
23. Owen VS, Rosgren B, Cherak SJ, et al. Adverse events associated with administration of vasopressor medications through a peripheral intravenous catheter: a systematic review and meta-analysis. *Crit Care* 2021; 25:146.
24. Prasanna N, Yamane D, Haridas N, et al. Safety and efficacy of vasopressor administration through midline catheters. *J Crit Care* 2021; 61:1–4.

25. Phentolamine mesylate. AHFS Drug Information, 2022. Bethesda, MD: American Society for Health System Pharmacists, 2022; 1436–1439.
26. Guyton AC, Hall JE. Medical Physiology. 10th ed. Philadelphia: W.B. Saunders, Co., 2000; 178.
27. Asfar P, Meziani F, Hamel JF, et al. High versus low blood-pressure target in patients with septic shock. *N Engl J Med* 2014; 370:1583–1593.

Vasopressor Drugs

28. Russell JA. Vasopressor therapy in critically ill patients with shock. *Intensive Care Med* 2019; 45:1503–1517.
29. Desairs P, Pinaud M, Bugnon D, Tasseau F. Norepinephrine therapy has no deleterious renal effects in human septic shock. *Crit Care Med* 1989; 17:426–429.
30. Anantasi N, Boyd JH, Walley KR, Russell JA. Serious adverse events associated with vasopressin and norepinephrine infusion in septic shock. *Crit Care Med* 2014; 42:1812–1820.
31. Stolk RF, van der Pasch E, Naumann F, et al. Norepinephrine dysregulated the immune response and compromises host defense during sepsis. *Am J Respir Crit Care Med* 2020; 202:830–842.
32. Epinephrine. AHFS Drug Information, 2022. Bethesda, MD: American Society of Health-System Pharmacists, 2022; 1424–1431.
33. De Backer D, Creteur J, Silva E, Vincent J-L. Effects of dopamine, norepinephrine, and epinephrine on the splanchnic circulation in septic shock: Which is best? *Crit Care Med* 2003; 31:1659–1667.
34. Levy B. Bench-to-bedside review: Is there a place for epinephrine in septic shock? *Crit Care* 2005; 9:561–565.
35. Dopamine hydrochloride. AHFS Drug Information, 2022. Bethesda, MD: American Society of Health-System Pharmacists, 2022; 1388–1392.
36. Avni T, Lador A, Lev S, et al. Vasopressors for the treatment of septic shock: systematic review and meta-analysis. *PLoS One* 2015; 10:e0129305.
37. Brath PC, MacGregor DA, Ford JG, Prielipp RC. Dopamine and intraocular pressure in critically ill patients. *Anesthesiology* 2000; 93:1398–1400.
38. Phenylephrine hydrochloride. AHFS Drug Information, 2022. Bethesda, MD: American Society of Health System Pharmacists, 2022; 1379–1384.
39. Morelli A, Ertmer C, Rehberg S, et al. Phenylephrine versus norepinephrine for initial hemodynamic support of patients with septic shock: a randomized, controlled trial. *Crit Care* 2008; 12:R143.
40. Treschan TA, Peters J. The vasopressin system: physiology and clinical strategies. *Anesthesiology* 2006; 105:599–612.
41. Polito A, Parisini E, Ricci Z, et al. Vasopressin for treatment of vasodilatory shock: an ESICM systematic review and meta-analysis. *Intensive Care Med* 2012; 38:9–19.
42. Wieruszewski PM, Wittwer ED, Kashani KB, et al. Angiotensin II infusion for shock. A multicenter study of postmarketing use. *Chest* 2021; 159:596–605.
43. Khanna A, English SW, Wang XS, et al. Angiotensin II for treatment of vasodilatory shock. *N Engl J Med* 2017; 377:419–430.
44. www.giapreza.com. Accessed June 23, 2023.
45. Chang TI. Impact of drugs on intradialytic hypotension: antihypertensive and vasoconstrictors. *Semin Dial* 2017; 30:532–536.
46. Santer P, Anstey MH, Patrocinio MD, et al. Effect of midodrine versus placebo on time to vasopressor discontinuation in patients with persistent hypotension in the intensive care unit (MIDAS): an international randomized clinical trial. *Intensive Care Med* 2020; 46:1884–1893.
47. Costa-Pinto R, Yong Z-T, Yanase F, et al. A pilot, feasibility, randomized controlled trial of midodrine as adjunctive vasopressor for low-dose vasopressor-dependent hypotension in intensive care patients: The MAVERIC study. *J Crit Care* 2022; 67:166–171.

Goals of Resuscitation

48. Fink MP. Cytopathic hypoxia. Mitochondrial dysfunction as mechanism contributing to organ dysfunction in sepsis. *Crit Care Clin* 2001; 17:219–237.
49. Ruggieri AJ, Levy RJ, Deutschman CS. Mitochondrial dysfunction and resuscitation in sepsis. *Crit Care Clin* 2010; 26:567–575.

A Final Word

50. Oriyanka P, Chang C-CH, Chawla LS, et al. Vasopressor-resistant hypotension, combination vasopressor therapy, and shock phenotypes in critically ill adults with vasodilatory shock. *Shock* 2022; 58:260–268.

Hemorrhagic Shock

It is a bad sign in acute illnesses when the extremities become cold.

Hippocrates *Aphorisms*, 400 B.C.

Trauma is the leading cause of death globally in the 25–49 year age group (1), and the most treatable cause of death in these cases is hemorrhagic shock. The resuscitation of traumatic blood loss presents many challenges (as described in this chapter), and one of these is the design of the circulatory system, which operates with a limited volume and a volume-sensitive stroke pump. This is an energy-efficient design that limits the workload of the heart, but it creates an intolerance to blood loss; e.g., whereas the lungs, liver, and kidneys can lose as much as 75% of their functional mass without a fatal outcome, loss of as little as 30% of the (limited) blood volume can be fatal. This intolerance to blood loss is the dominant concern in the bleeding patient.

This chapter describes the physiology and classification of acute blood loss, and presents the relevant features and strategies involved in the resuscitation of hemorrhagic shock, with an emphasis on life-threatening blood loss from traumatic injuries.

ACUTE BLOOD LOSS

The normal blood volume in adults is estimated at 66 mL/kg (lean body weight) for males, and 60 mL/kg for females (2). For an adult male with a lean body weight of 75 kg (165 lbs), the estimated blood volume is $66 \times 75 = 5$ liters, while an adult female with a lean body weight of 60 kg (132 lbs) has an estimated blood volume of $60 \times 60 = 3.6$ liters.

Compensatory Responses

Acute blood loss triggers the following compensatory responses, which help to mitigate the adverse consequences of the volume loss (3,4).

- . The decrease in blood pressure promotes the movement of interstitial fluid into the bloodstream. This *transcapillary refill* can add as much as one liter to the plasma volume, but it leaves an interstitial fluid deficit.

- . There is a physiological stress response that involves activation of the sympathetic nervous system, which promotes tachycardia and peripheral vasoconstriction, both aimed at preserving tissue perfusion.
- . The stress response also stimulates the release of antidiuretic hormone (ADH) from the posterior pituitary gland, which counteracts the volume loss by promoting the renal retention of water.
- . The decrease in renal perfusion activates the renin–angiotensin–aldosterone system. Angiotensin promotes widespread vasoconstriction, which helps to support tissue perfusion pressure, while aldosterone promotes renal sodium retention, which provides support for the extracellular volume.

Progressive Blood Loss

Acute blood loss can be described with the following classification system (5), which is also outlined in [Table 15.1](#).

TABLE 15.1 Classification System for Acute Blood Loss				
	Class I	Class II	Class III	Class IV
Volume Deficit	<15%	15–30%	31–40%	>40%
Volume Loss	<10 mL/kg	10–20 mL/kg	21–30 mL/kg	>30 mL/kg
Heart Rate	↔	↑	↑↑	↑↑ / ↓
Blood Pressure	↔	↔ / ↓	↓	↓↓
Urine Output	↔	↔ / ↓	↓	↓↓
Plasma Lactate	↔	↔	↑	↑↑
Interpretation	Asymptomatic Phase	Compensated Phase	Shock Phase	Advanced Shock
Resuscitation	None	Crystalloid Fluids	Blood	Massive Transfusion

Adapted from Reference 5.

CLASS I: Loss of <15% of the blood volume (or <10 mL/kg). The volume loss in this case (which is typically less than one liter) is fully restored by transcapillary refill, so blood volume is maintained, clinical findings are minimal or absent, and volume resuscitation is not necessary (3).

CLASS II: Loss of 15–30% of the blood volume (or 10–20 mL/kg). This represents the compensated phase of hypovolemia, where blood pressure is maintained by systemic vasoconstriction (4). Urine output may begin to fall, but does not reach the level of oliguria (<0.5 mL/kg/hr). The vasoconstrictor response is most intense in the splanchnic circulation, and splanchnic hypoperfusion can lead to disruption of the intestinal mucosa and invasion of the bloodstream with enteric pathogens (6).

CLASS III: Loss of 31–40% of the blood volume (or 21–30 mL/kg). This marks the onset of decompensated blood loss or *hemorrhagic shock*, where the vasoconstrictor response is no longer able to sustain blood pressure and organ perfusion. The clinical consequences can include supine hypotension, signs of impaired organ perfusion (oliguria, altered mentation, cool extremities) and evidence of anaerobic metabolism (i.e., lactate accumulation in blood).

CLASS IV: Loss of >40% of blood volume (or >30 mL/kg). This degree of blood loss results in profound hemorrhagic shock, which may be irreversible. Clinical manifestations include multiorgan failure and severe metabolic (lactic) acidosis. This category includes *massive blood loss*, which is described later in the chapter.

Hemoglobin and Hematocrit

The use of the hemoglobin concentration and hematocrit (H & H) to evaluate the presence and severity of acute blood loss is both common and inappropriate. Changes in hematocrit show a poor correlation with blood volume deficits and erythrocyte deficits in acute hemorrhage (7), and the reason for this discrepancy is demonstrated in Figure 15.1. Acute blood loss involves the loss of whole blood, which results in proportional decreases in the volume of plasma and erythrocytes. As a result, acute blood loss results in a decrease in blood volume, but not a decrease in the H & H. The H & H will eventually decline as a result of transcapillary refill, and the compensatory increase in antidiuretic hormone and aldosterone activity mentioned previously, but this effect is not apparent for 8–12 hours, and it can take a few days to become fully established.

The right side of Figure 15.1 shows how the resuscitation fluid influences the hematocrit. Infusion of isotonic saline increases the plasma volume but not the red cell volume, resulting in a dilutional decrease in the hematocrit (8). On the other hand, resuscitation with whole blood adds to both plasma and red cell volumes proportionally, so there is no change in hematocrit. These examples demonstrate that *in the early hours after acute blood loss, the hemoglobin and hematocrit are reflections of the resuscitation effort (the type and volume of fluids infused), and not the extent of blood loss.*

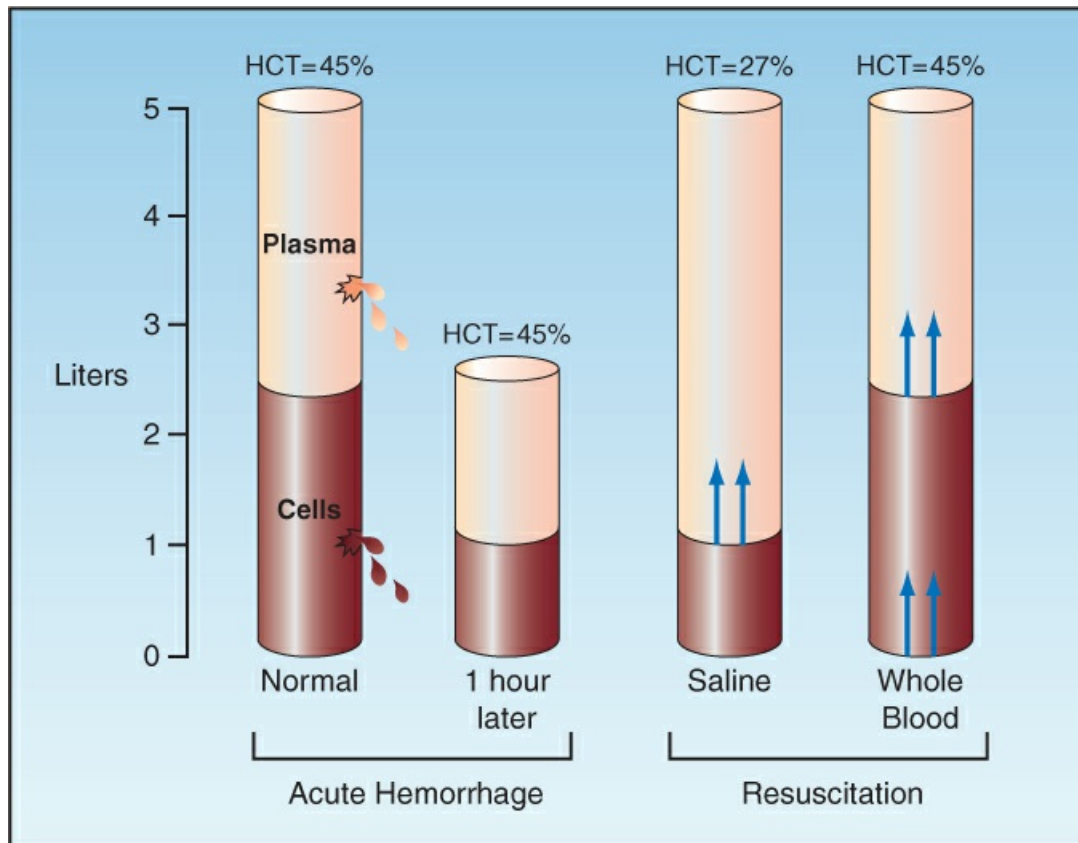


FIGURE 15.1 Influence of acute hemorrhage and fluid resuscitation on blood volume and hematocrit. See text for explanation.

RESUSCITATION BASICS

The Catheters

The determinants of flow through vascular catheters are described in [Chapter 1](#) using the Hagen-Poiseuille equation, which is shown below (9).

$$Q = \Delta P \times (\pi r^4 / 8 \mu L) \quad (15.1)$$

This equation states that the steady or laminar flow (Q) in a rigid tube (catheter) is directly related to the pressure gradient along the tube ($\Delta P = P_{in} - P_{out}$) and the fourth power of the radius of the tube (r^4), and is inversely related to the length of the tube (L) and the viscosity of the fluid (μ). These relationships indicate that *short, large-bore peripheral vein catheters are much better suited for rapid infusions than the longer central venous catheters*. This is demonstrated in [Figure 15.2](#), which shows that for a given gauge size, the gravity-driven flow of water through a 1.2-inch peripheral vein catheter is about 5 times greater than the flow through an 8-inch central venous catheter. Also note that increasing the bore size of a catheter from 18 gauge to 16 gauge results in a two-fold increase in flow rate.

Rapid Infusion Catheters

Very large bore peripheral catheters are available in 7 French and 9 French sizes (equivalent to 13 gauge and 11 gauge sizes, respectively) and are about 2 inches in length. These *rapid infusion catheters* are typically used with pressurized infusion systems (see later), and they can deliver up to one liter per minute (10).

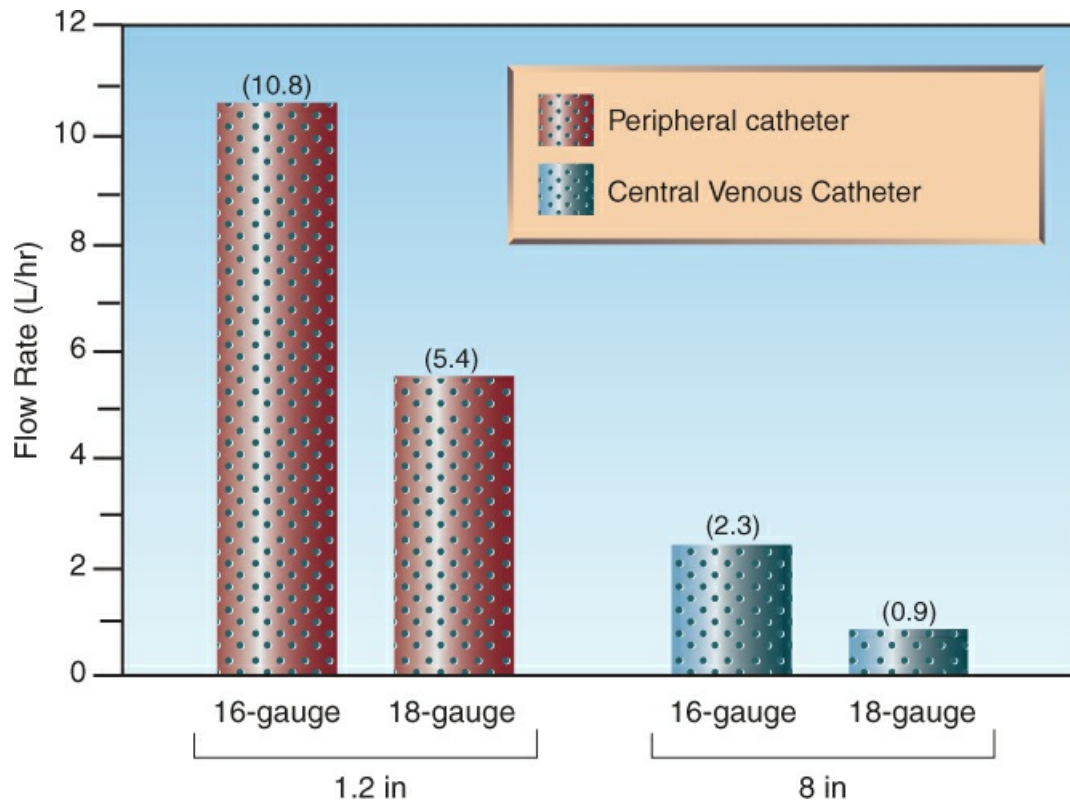


FIGURE 15.2 The influence of catheter dimensions on the gravity-driven flow of water. Data from www.emnote.org (for peripheral catheters) and www.teleflex.com (for central venous catheters).

Introducer Catheters

Large-bore introducer catheters are used as conduits for multilumen catheters (see Figure 8.1), but they can also be used as stand-alone infusion devices. They are available in 8.5 French and 9 French sizes (equivalent to about 11 gauge), and can deliver up to one liter per minute when used with pressurized infusion systems (10). These catheters are about 10 cm (4 inches) in length, and are inserted into large central veins (which make them less appealing than the rapid infusion catheters). Some introducer catheters have an additional side infusion port (see Figure 8.1), but the flow capacity of this port is only 25% of the flow capacity of the catheter (11), so it should be bypassed for rapid infusion rates.

Resuscitation Fluids

The different types of resuscitation fluids for acute blood loss are shown in Table 15.2. Each of these fluids (except whole blood) has been described in detail elsewhere in the book, and the following is only a brief summation.

TABLE 15.2 Resuscitation Fluids for Blood Loss		
Type of Fluid	Products	Principal Effect
Colloid Fluid	Albumin (5%, 25%), Hydroxyethyl Starches, Dextrans	Expands the plasma volume
Crystalloid Fluid	Isotonic Saline, Ringer's lactate, Normosol, Plasma-Lyte	Expands the extracellular volume
RBC Concentrate	Packed RBCs	Increases the O ₂ content of blood
Plasma	Fresh Frozen Plasma, Liquid Plasma	Provides procoagulant proteins
Plasma Precipitate	Cryoprecipitate, Fibrinogen Concentrate	Increases fibrinogen levels
Platelet Concentrate	Multiple-Donor Platelets, Single-Donor Platelets	Increases circulating platelets
Whole Blood	Type-Specific Blood Group O Blood	All of the above

Asanguinous Fluids

Asanguinous fluids (colloids and crystalloids) are used in the resuscitation of moderate (Class II) blood loss, and in the initial stage of resuscitation for hemorrhagic shock. Colloid fluids are much more effective for promoting the cardiac output (see [Figure 15.3](#)), because they expand the plasma volume more than crystalloid fluids, as explained in [Chapter 10](#). However, despite the superiority of colloid fluids for expanding the plasma volume, crystalloid fluids have been the preferred fluid for the resuscitation of acute blood loss (see next).

CRYSTALLOIDS AND BLOOD LOSS: The perceived value of crystalloid fluids in hemorrhagic shock can be traced back to experimental studies in the 1960s. Using an animal model of hemorrhagic shock, these studies showed that few animals survived when shed blood, or shed blood plus plasma, was reinfused, but survival markedly improved when a crystalloid fluid (Ringer's lactate) was added to shed blood replacement ([13](#)). These same studies also showed that there was a disproportionate (40%) decrease in the volume of interstitial fluid in hemorrhagic shock, and further, that this volume was restored only when the crystalloid fluid was added to the shed blood replacement. These observations implied that an interstitial fluid deficit was a critical factor in the inability to survive hemorrhagic shock, and the crystalloid fluid resuscitation was needed to correct this deficit (and improve survival).

The consensus view at the present time is that fatal outcomes in hemorrhagic shock are the result of persistent bleeding and an unrelenting coagulopathy (described later in the chapter), and that the aggressive use of crystalloid fluids can aggravate ongoing blood loss by promoting a dilutional decrease in coagulation factors. As a result, the current emphasis is on limiting the use

of crystalloid fluids in the resuscitation of life-threatening hemorrhage (see “damage control resuscitation” later in the chapter).

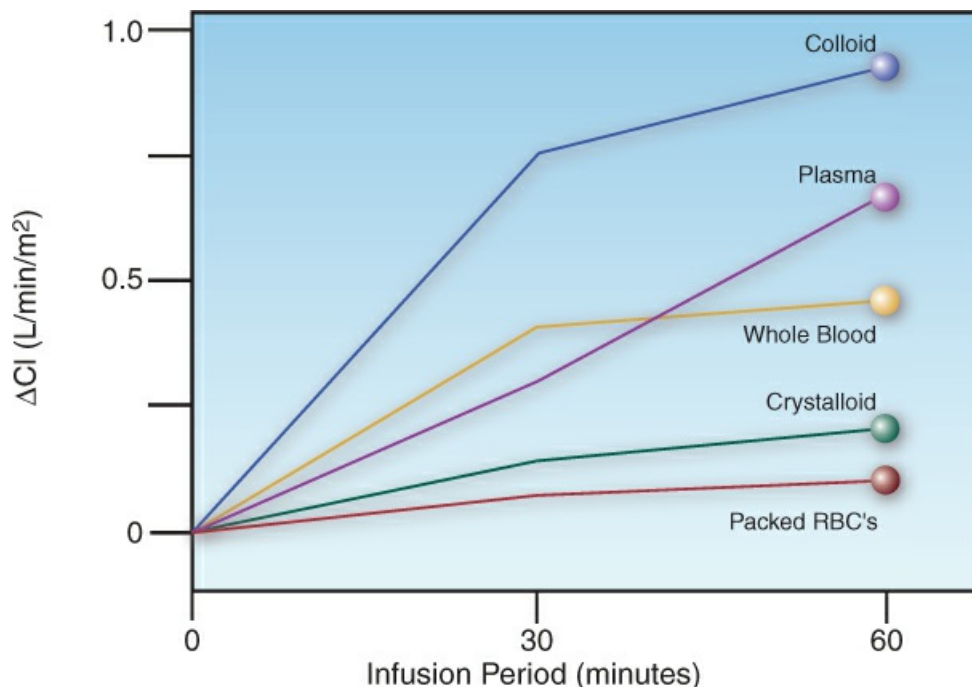


FIGURE 15.3 The influence of a one-hour infusion of selected resuscitation fluids on the cardiac index (CI). The colloid is 10% dextran-40, and the crystalloid is Ringer's lactate. Infusion volumes are roughly equivalent (500 mL), except for the crystalloid (1 liter). Data from Reference 12.

Erythrocyte Concentrates

Erythrocyte concentrates, better known as “packed red blood cells” are produced by centrifugation of donor blood. Each unit has a hematocrit of about 60% and a volume of about 350 mL (see [Table 12.3](#)). Packed RBCs do not promote cardiac output (see [Figure 15.3](#)) because of their high viscosity, and they are used only to increase the oxygen carrying capacity of blood. (See [Chapter 12](#) for more on packed RBCs.)

Plasma Products

Plasma is stored at -18°C (-0.4°F), and must be thawed before using. This “fresh frozen plasma” (FFP) is available in aliquots (units) of about 230 mL, and is used as a source of coagulation factors, and as a volume expander in massive transfusion protocols (see later). FFP has two disadvantages: i.e., the time required for thawing can lead to delays in resuscitation, and there is a limited shelf life after thawing, which can lead to wastage. These disadvantages are not shared by *liquid plasma*, which is not frozen, and can be stored at $34^{\circ}\text{--}43^{\circ}\text{F}$ for up to 26 days. Liquid plasma has proven equivalent in efficacy to FFP while also reducing wastage ([14](#)). (See [Chapter 13](#) for more on plasma transfusions.)

Cryoprecipitate

Cryoprecipitate is a frozen plasma precipitate that is rich in fibrinogen and Factor VIII. It has a

volume of about 10–15 mL, and is used to raise plasma fibrinogen levels (15). Cryoprecipitate has the same disadvantages as FFP: i.e., must be thawed before use, and has a limited shelf life after thawing. Other disadvantages include a variable fibrinogen content, and an increased risk of transmitted infections. These problems are not shared by *fibrinogen concentrates*, which have a consistent fibrinogen content (20 g/L), do not require thawing, and have no risk of transmitted infections (16).

Whole Blood

Whole blood is considered the ideal replacement fluid for acute blood loss, and it can be stored at 1°–6° C (34°–43° F) for up to 35 days (17). However, whole blood is not readily available, because of the standard practice of separating donor blood into its components, which optimizes its utilization.

Fresh whole blood has been used during the recent military conflicts in the Middle East, which was made possible by the use of “walking blood banks” (i.e., fellow soldiers), and it showed a survival benefit when compared to component therapy (17). As a result of this experience, cold-stored whole blood has been evaluated in a civilian setting, but it has not shown the same survival benefit as it did in the military studies (18,19). However, there is evidence of a survival benefit when whole blood is combined with component therapy (19), and it may be that whole blood will fill the gap when there is a limited supply of blood components.

LOW-TITER GROUP O BLOOD: Group O blood is considered universal donor blood, and is used when type-specific blood is not immediately available. However, transfusion reactions have occurred with Group O blood that has exceptionally high titers of anti-A and anti-B antibodies (20). “Low-titer” Group O blood has low levels of anti-A and anti-B antibodies, and is considered a safer alternative to Group O blood. The experience with low-titer Group O blood has been similar to whole blood; i.e., a survival benefit has been demonstrated in military studies, but not in the civilian population (21).

Rapid Infusion Systems

Rapid blood replacement is often necessary in massive hemorrhage, and rapid infusion systems are available that can deliver warmed blood products at rates up to 1 liter per minute (22). These systems use a roller pump to create pressures up to 300 mm Hg to deliver a preselected infusion rate. Heating coils are also used to deliver the infusate at body temperature, which eliminates the risk of hypothermia from infusing room-temperature fluids.

Pressurized infusion systems are especially helpful in transfusing erythrocyte concentrates (packed RBCs), which are highly viscous and do not flow readily. This is demonstrated in Table 15.3, which shows that the gravity-driven infusion of packed RBCs is sluggish, and is improved by dilution with saline (23). However, rates of blood replacement can be 5 L/hr (83 mL/min) or higher in life-threatening hemorrhage, and these rates can only be reached with pressurized infusions. Concerns about hemolysis with rapid infusion systems have not been validated (24).

TABLE 15.3 Infusion of Packed RBCs: Influence of Dilution, Pressure, and Catheter Size

--	--

	Maximum Flow (mL/min)		
	18 ga	16 ga	7 Fr RIC
	1.2 in	1.2 in	2.0 in
1 Unit PRBCs – No Dilution Gravity-Driven Flow	5	7	—
1 Unit PRBCs + 100 mL Saline Gravity-Driven Flow	39	49	—
1 Unit PRBCs + 1 Unit FFP Pressurized Flow (300 mm Hg)	183	448	1,000

Data from References 24 and 25. RIC = Rapid infusion catheter.

RESUSCITATION STRATEGIES

There are two fundamental goals in the bleeding patient. The first goal is to support the delivery of oxygen to tissues by promoting cardiac output and maintaining an adequate hemoglobin concentration. The second (and equally important) goal is “source control” to stop the bleeding, as continued bleeding not only depletes resources, it increases the risk of a life-threatening coagulopathy (see later).

TABLE 15.4 Asanguinous Resuscitation Volumes for Class II Hemorrhage

Steps	Methods
1. Estimate normal plasma volume (PV)	PV = 40 mL/kg (males)* = 36 mL/kg (females)
2. Estimate % loss of plasma volume	Class II: 15–30% Loss
3. Calculate plasma volume deficit (PVD)	PVD = PV x % Loss
4. Estimate resuscitation volume (RV)	RV = PVD x 1 (colloids) = PVD x 3 (crystalloids)

* Estimated plasma volumes from Reference 2.

Hemorrhage Without Shock

Blood loss that is accompanied by a decrease in blood pressure or urine output, but without evidence of shock (e.g., plasma lactate is not elevated) can be managed initially with asanguinous fluids. (This is equivalent to Class II blood loss presented earlier.) The resuscitation volume can be estimated as shown in [Table 15.4](#). Note that the volume for crystalloid fluids is 3 times the estimated loss of plasma volume, which is a conservative estimate (since only 20–25% of crystalloid fluids expand the plasma volume).

When the plasma volume is repleted, packed RBCs may be needed to attain a target Hb of 7–8 g/dL (see [Chapter 12](#)), FFP may be needed if the INR is >1.5 ([25](#)), and platelet transfusions are indicated if the platelet count falls below 50,000/ μ L (see [Chapter 13](#)). If bleeding continues, then the resuscitation strategy described next should be considered.

Hemorrhagic Shock

The resuscitation of hemorrhagic shock described here follows the principles of “damage control resuscitation” (26), which is typically used for massive blood loss (e.g., infusion of 4 units of RBCs in one hour), but will benefit all cases of life-threatening hemorrhage. The basic tenets of damage control resuscitation include timely source control, permissive hypotension, restricting the infusion of crystalloid fluids, and resuscitation with whole blood, or its equivalent in component therapy.

Permissive Hypotension

The impetus for permissive hypotension comes from studies of trauma-related hemorrhage, which have shown that aggressive fluid resuscitation can exacerbate bleeding before the hemorrhage is controlled (27). This led to an emphasis on reducing fluid resuscitation to allow blood pressures to remain lower than normal (e.g., systolic BP 80–90 mm Hg or mean BP of 50 mm Hg) until the bleeding is controlled. This strategy has been shown to reduce transfusion requirements and the risk of a dilutional coagulopathy (28,29), but the effect on mortality rates has been inconsistent (27–29). The major issue with this strategy is ensuring that the low blood pressure is not promoting dangerous tissue hypoperfusion, so monitoring plasma lactate levels and markers of organ perfusion, such as urine output, is essential (as it should be in all cases of hemorrhagic shock).

Coagulopathy

One of the principal fears in the bleeding patient is persistent blood loss, which can be the result of inadequate source control and/or a coagulopathy. The latter process is often the result of acidosis (which inhibits coagulation factors at a pH <7.2), hypothermia (which decreases platelet adherence), or the infusion of asanguinous fluids (which has a dilutional effect on coagulation factors) (30). However trauma victims can have a coagulopathy that is independent of the traditional sources, and this “trauma-induced coagulopathy” (TIC) has a negative impact on outcomes. Studies of combat casualties in the Middle East showed that TIC was present on admission in 40% of the cases, and the presence of TIC created a 6-fold increase in the mortality rate (31).

The major culprit in TIC is activation of protein C as a result of endothelial injury (27). The activated protein C not only deactivates factors V and VIII (which inhibits clot formation), it also triggers fibrinolysis (which promotes clot dissolution, and leads to a consumptive decrease in fibrinogen levels) (30). Platelet dysfunction may also play a role in TIC (32).

Hemostatic Resuscitation

The basic elements of hemostatic resuscitation are outlined in [Figure 15.4](#) (33). The most important element is the infusion of RBCs, plasma, and platelets in proportions that mimic whole blood replacement. This 1:1:1 replacement strategy can be achieved by infusing one unit of FFP for every unit of packed RBCs, and adding one platelet pack (multidonor or single-donor platelets) for every 5 units of PRBCs and FFP. (Remember that platelet transfusions are equivalent to the platelets from 5–6 donors.) This approach not only optimizes the replacement of clotting elements (coagulation factors and platelets), it also minimizes the risk of a dilutional coagulopathy from the infusion of crystalloid fluids. Infusions should be warmed to body temperature to avoid hypothermia.

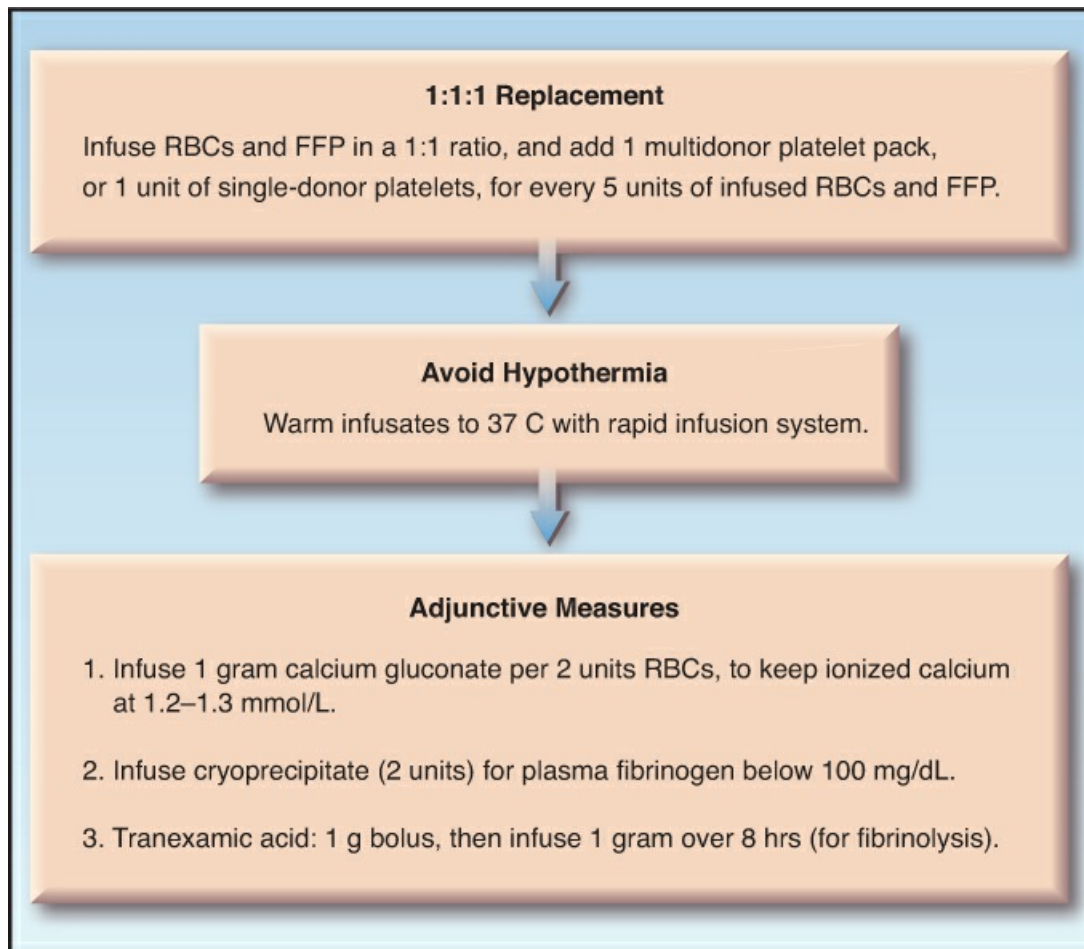


FIGURE 15.4 The principal components of hemostatic resuscitation. See text for explanation.

ADJUNCTIVE MEASURES: Other measures that promote hemostasis include calcium supplementation to maintain an ionized calcium of 1.2–1.3 mmol/L (4.6–6 mg/dL), cryoprecipitate (two units) for a plasma fibrinogen level <100 mg/dL, and empiric treatment for fibrinolysis with tranexamic acid (1 gram IV as a bolus dose, followed by infusion of 1 gram over 8 hours) (33).

END-POINTS: The end-points of hemostatic resuscitation include a plasma fibrinogen level >100 mg/dL, an INR <1.5, an activated PTT <1.5 times normal, and a platelet count of $\geq 100,000/\mu\text{L}$ (30). However, these conventional measures are not well suited for detecting the hemostatic defects in TIC, and “viscoelastic assays” like *thromboelastography* have proven to be more sensitive methods of monitoring hemostasis in trauma victims (see next).

Thromboelastography

Thromboelastography (TEG) provides information on clot initiation, progression, strength, and lysis from whole blood samples at body temperature. This is accomplished by placing a sample of blood in a cylindrical cup that has a central pin suspended in the sample. The cup is then rotated in an alternating clockwise and counterclockwise direction, and the stationary pin records the torque produced as the blood coagulates and then lyses. (An alternative method known as

rotational thromboelastometry or ROTEM, which uses a stationary cup and a rotating pin, is not described here.)

A typical TEG tracing, which is shown in [Figure 15.5](#), is a record of the resistance to oscillations in the sample (used as a measure of clot firmness) over time. The symmetrical curves are produced by the alternating clockwise and counterclockwise rotation of the blood sample. The relevant measures are as follows (34):

- . R is the “reaction time”, or is the time to initiation of clot formation. The normal R time is 5–10 minutes, and is prolonged by defects in the coagulation cascade, including those produced by all anticoagulants except warfarin (i.e., the INR is the only reliable measure of warfarin anticoagulation).
- . k is the “kinetic time”, and is the time from clot initiation to a resistance of 20 mm. The normal k time is 1–3 minutes.
- . The α angle represents the early rate of clot formation, and is dependent on fibrin polymerization. A normal α angle is 53° – 72° .
- . MA is the maximum amplitude, and is a measure of clot strength. A normal amplitude is 50–70 mm, and is a reflection of fibrinogen cross-bridging of platelets.
- . LY30 is the percentage of clot lysis that occurs from maximal amplitude (MA) to 30 minutes. This is a measure of fibrinolysis, and the normal range is 2.3–5.7%.

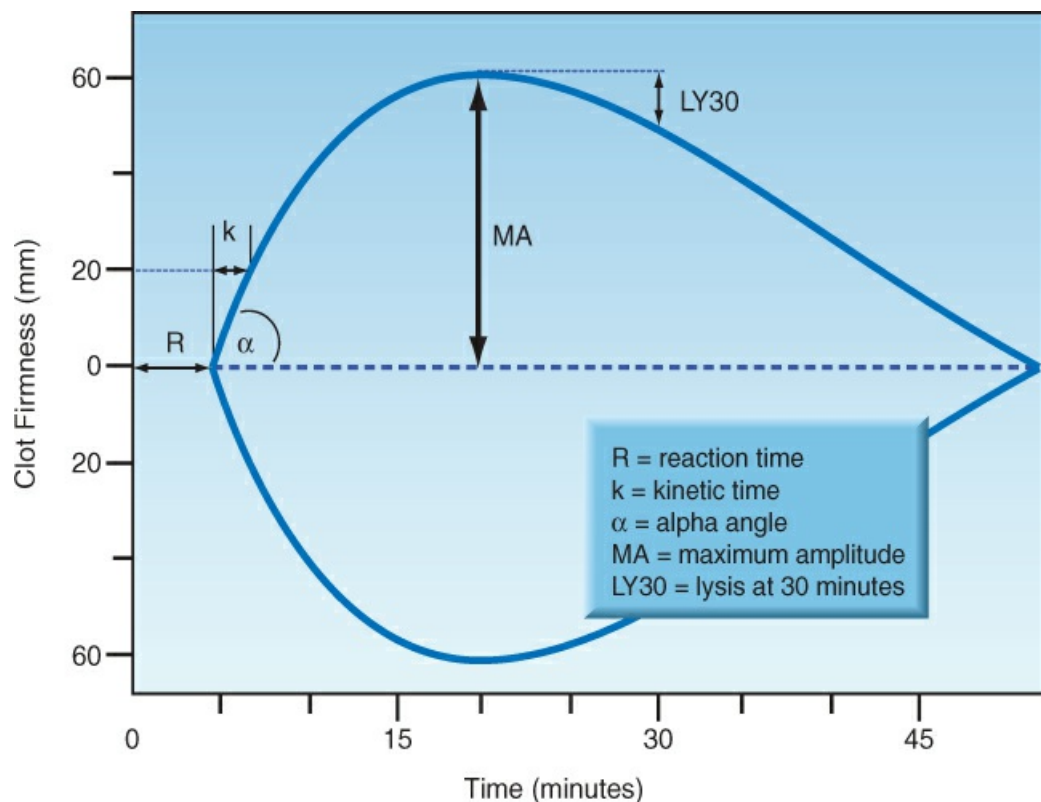


FIGURE 15.5 A thromboelastography tracing with identified components. See text for explanation.

TEG-Guided Therapy

The TEG measures just described can be used to guide hemostatic therapy as shown in [Table 15.5](#). Clinical studies involving both trauma victims and postoperative patients with life-threatening hemorrhage ([35,36](#)) have shown that hemostatic management guided by TEG improves survival and reduces the use of blood products.

TABLE 15.5 Interventions Based on Thromboelastography		
Measure	Abnormal Value	Intervention
Reaction (R) Time	>9 minutes	Transfuse plasma, or reverse anticoagulants (except warfarin)
Kinetic (k) Time	>2.5 minutes	Transfuse plasma
Alpha (α) Angle	<65°	Cryoprecipitate or fibrinogen concentrate
MA (max amplitude)	<55 mm	Transfuse platelets
LY30 (lysis in 30 min)	>3%	Tranexamic acid

From Reference 34.

Goals of Resuscitation

The overall goals of resuscitation for hemorrhagic shock are summarized in [Figure 15.6](#). Many of the tissue perfusion and oxygenation goals have been presented elsewhere in the book, while the hemostasis goals have just been described. Unfortunately, achieving these goals does not ensure a satisfactory outcome, as described next.

POSTRESUSCITATION INJURY

An apparent successful resuscitation of hemorrhagic shock can be followed in 48–72 hours by progressive multiorgan failure ([37](#)). The earliest manifestation is progressive respiratory dysfunction, which is followed within days by progressive dysfunction of the kidneys, liver, heart, and central nervous system. The mortality rate is determined by the number of organs involved, and averages 50–60% ([37](#)).

Pathophysiology

Postresuscitation injury is a form of *reperfusion injury* ([38](#)) that is believed to originate in the splanchnic circulation, where reperfusion of ischemic bowel releases proinflammatory cytokines that gain access to the systemic circulation and incite a systemic inflammatory response that is responsible for the multiorgan damage.

Several factors predispose to postresuscitation injury, including the time required to reverse the shock, the volume of crystalloid resuscitation (which promotes bowel edema and predisposes to the abdominal compartment syndrome), and the number of units of RBCs transfused (>6 units in 12 hrs) ([38](#)). Infection of bowel origin may be involved if the onset of multiorgan failure is more than 3 days after the resuscitation ([38](#)).

Management

There is no specific therapy for postresuscitation injury, and preventive measures such as rapid reversal of the shock state, and limiting the infusion of crystalloid fluids and RBC products (if possible) may be helpful. In late-onset multiorgan failure (onset >72 hrs after resuscitation), recognition and prompt treatment of an underlying sepsis is essential.

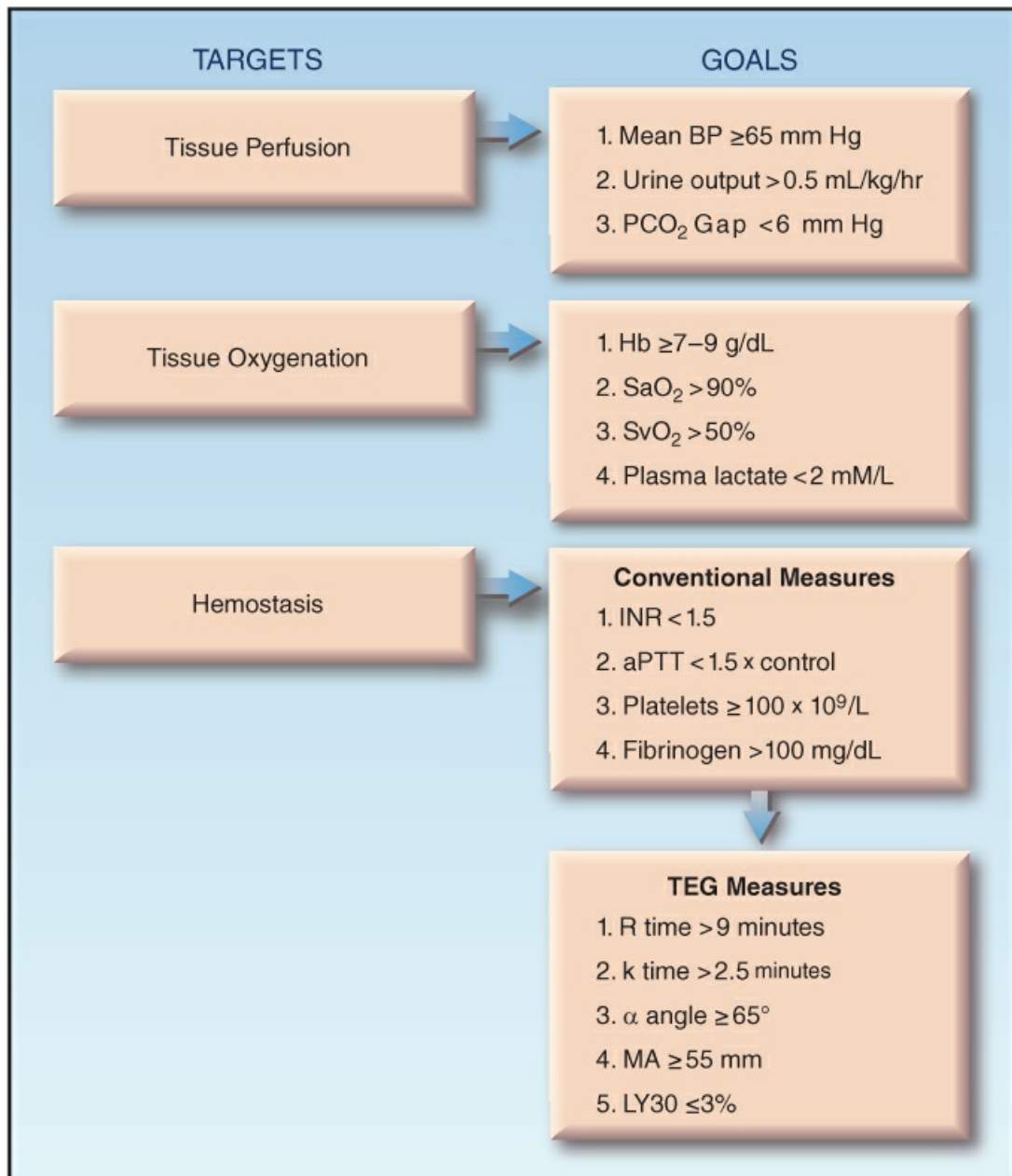


FIGURE 15.6 A summation of the goals of resuscitation in hemorrhagic shock.

A FINAL WORD

A Summation

The resuscitation of hemorrhagic shock should incorporate the following elements (in addition to timely source control):

- . Use short, large-bore catheters, and avoid hypothermia by warming infusates to 37° C (e.g., with a rapid infusion system).
- . Restrict the use of crystalloid fluids to 1 or 2 liters before using blood products (if possible).
- . Replace blood components in a ratio that will mimic whole blood replacement, and consider the adjunctive measures shown in [Figure 15.4](#).
- . Consider “permissive hypotension” until the bleeding is controlled. The target BP in this approach is a systolic pressure of 80–90 mm Hg or a mean pressure of 50 mm Hg, which is maintained as long as there is evidence of adequate tissue perfusion (e.g., plasma lactate <2 mmol/L).
- . If possible, use thromboelastography (or rotational thromboelastometry) to guide hemostatic management (see [Figure 15.5](#) and [Table 15.5](#)).

And if the resuscitation is successful, remember to remain vigilant for the appearance of postresuscitation organ injury and sepsis of bowel origin.

References

1. GBD 2019 Diseases and Injuries Collaborators. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet* 2020; 396:1204–1222.

Acute Blood Loss

2. Walker RH (ed). *Technical Manual of the American Association of Blood Banks*. 10th ed., Arlington, VA: American Association of Blood Banks, 1990; 650.
3. Moore FD. Effects of hemorrhage on body composition. *New Engl J Med* 1965; 273:567–577.
4. Schadt JC, Ludbrook J. Hemodynamic and neurohumoral responses to acute hypovolemia in conscious animals. *Am J Physiol* 1991; 260:H305–318.
5. ATLS, Advanced Trauma Life Support. *Student Course Manual*. 10th ed. Chicago, IL: American College of Surgeons, 2018; 43–61.
6. Fiddian-Green RG. Studies in splanchnic ischemia and multiple organ failure. In: Marston A, Bulkley GB, Fiddian-Green RG, Haglund UH, eds. *Splanchnic ischemia and multiple organ failure*. St. Louis, CV Mosby, 1989; 349–364.
7. Cordts PR, LaMorte WW, Fisher JB, et al. Poor predictive value of hematocrit and hemodynamic parameters for erythrocyte deficits after extensive vascular operations. *Surg Gynecol Obstet* 1992; 175:243–248.
8. Stamler KD. Effect of crystalloid infusion on hematocrit in nonbleeding patients, with applications to clinical traumatology. *Ann Emerg Med* 1989; 18:747–749.

Resuscitation Basics

9. Chien S, Usami S, Skalak R. Blood flow in small tubes. In: Renkin EM, Michel CC (eds). *Handbook of Physiology*. Section 2: The cardiovascular system. Volume IV. The microcirculation. Bethesda: American Physiological Society, 1984; 217–249.
10. Milne A, Teng JJ, Vargas A, et al. Performance assessment of intravenous catheters for massive transfusion: A pragmatic in vitro study. *Transfusion* 2021; 61:1721–1728.
11. Hyman SA, Smith DW, England R, et al. Pulmonary artery catheter introducers: Do the component parts affect flow rate? *Anesth Analg* 1991; 73:573–575.
12. Shoemaker WC. Relationship of oxygen transport patterns to the pathophysiology and therapy of shock states. *Intensive Care Med* 1987; 213:230–243.
13. Shires T, Coln D, Carrico J, Lightfoot S. Fluid therapy in hemorrhagic shock. *Arch Surg* 1964; 88:688–693.
14. Smith S, Dewey E, VanSandt M, Schreiber M. Liquid plasma reduces waste and health care expenses compared with thawed plasma at a level 1 trauma center. *J Trauma Acute Care Surg* 2020; 89:867–870.
15. Nair PM, Rendo MJ, Reddoch-Cardenas KM, et al. Recent advances in use of fresh frozen plasma, cryoprecipitate,

immunoglobulins, and clotting factors for transfusion support in patients with hematologic disease. *Semin Hematol* 2020; 57:73–82.

16. Grottke O, Mallaiah S, Karkouti K, et al. Fibrinogen supplementation and its indications. *Semin Thromb Hemost* 2020; 46:38–49.
17. McCoy CC, Brenner M, Duchesne J, et al. Back to the future: whole blood resuscitation of the severely injured trauma patient. *Shock* 2021; 56(Suppl 1):9–15.
18. Malkin M, Nevo A, Brundage SI, Schreiber M. Effectiveness and safety of whole blood compared to balanced blood components in resuscitation of hemorrhaging trauma patients—a systematic review. *Injury* 2021; 52:182–188.
19. Ngatuvai M, Zagales I, Sauder M, et al. Outcomes of transfusion with whole blood, component therapy, or both in adult civilian trauma patients: a systematic review and meta-analysis. *J Surg Res* 2023; 287:193–201.
20. Berseus O, Boman K, Nessen SC, Westerberg LA. Risks of hemolysis due to anti-A and anti-B caused by the transfusion of blood or blood components containing ABO-incompatible plasma. *Transfusion*. 2013; 53(Suppl 1):114S–123S.
21. Dilday J, Lewis MR. Transfusion management in the trauma patient. *Curr Opin Crit Care* 2022; 28:725–731.
22. Milne A, Teng JJ, Vargas A, et al. Performance assessment of intravenous catheters for massive transfusion: a pragmatic in vitro study. *Transfusion* 2021; 61:1721–1728.
23. de la Roche MRP, Gauthier L. Rapid transfusion of packed red blood cells: effects of dilution, pressure, and catheter size. *Ann Emerg Med* 1993; 22:1551–1555.
24. Kim P, Chin-Yee I, Eckert K, et al. Hemolysis with rapid transfusion systems in the trauma setting. *Can J Surg* 2004; 47:295–297.

Resuscitation Strategies

25. Stainsby D, MacLennan S, Thomas D, et al, for the British Committee for Standards in Hematology. Guidelines on the management of massive blood loss. *Br J Haematol* 2006; 135:634–641.
26. Cannon J, Khan M, Raja A, et al. Damage control resuscitation in patients with severe traumatic hemorrhage: a practice management guideline from the Eastern Association for the Surgery of Trauma. *J Trauma Acute Care Surg* 2017; 82:605–17.
27. Kudo D, Yoshida Y, Kushimoto S. Permissive hypotension/hypotensive resuscitation and restricted/controlled resuscitation in patients with severe trauma. *J Intensive Care* 2017; 5:11.
28. Bickell WH, Wall MJ Jr, Pepe PE, et al. Immediate versus delayed fluid resuscitation for hypotensive patients with penetrating torso injuries. *N Engl J Med* 1994; 331:1105–1109.
29. Morrison CA, Carrick M, Norman MA, et al. Hypotensive resuscitation strategy reduces transfusion requirements and severe postoperative coagulopathy in trauma patients with hemorrhagic shock: preliminary results of a randomized controlled trial. *J Trauma* 2011; 70:652–663.
30. Cohen MJ, Christie SA. Coagulopathy of trauma. *Crit Care Clin* 2017; 33:101–118.
31. Niles SE, McLaughlin DF, Perkins JG, et al. Increased mortality associated with the early coagulopathy of trauma in combat casualties. *J Trauma* 2008; 64:1459–1463.
32. Davenport RA, Brohi K. Coagulopathy in trauma patients: importance of thrombocyte function? *Curr Opin Anaesthesiol* 2009; 22:261–266.
33. Napolitano LM. Hemostatic defects in massive transfusion: an update and treatment recommendations. *Exp Rev Hematol* 2021; 14:219–239.
34. Brill JB, Brenner M, Duchesne J, et al. The role of TEG in damage control resuscitation. *Shock* 2021; 56(Suppl 1):52–61.
35. Wikkelsø A, Wetterslev J, Møller AM, Afshari A. Thromboelastography (TEG) or rotational thromboelastometry (ROTEM) to monitor haemostatic treatment in bleeding patients: a systematic review with meta-analysis and trial sequential analysis. *Anaesthesia* 2017; 72:519–531.
36. Gonzalez E, Moore EE, Moore HB, et al. Goal-directed hemostatic resuscitation of trauma-induced coagulopathy: a pragmatic randomized clinical trial comparing a viscoelastic assay to conventional coagulation assays. *Ann Surg* 2016; 263:1051–1059.

Postresuscitation Injury

37. Dewar D, Moore FA, Moore EE, Balogh Z. Postinjury multiorgan failure. *Injury* 2009; 40:912–918.
38. Eltzschig HK, Collard CD. Vascular ischaemia and reperfusion injury. *Br Med Bull* 2004; 70:71–86.

Chapter 16

Cardiogenic Shock

When is a piece of matter said to be alive? When it goes on “doing something,” moving, exchanging material with its environment.

Erwin Schrödinger ([a](#))

The human organism goes on exchanging material with its environment by virtue of a circulatory system equipped with an automatic stroke pump that averages about 100,000 stroke cycles daily, and pumps a daily volume of about 8,000 liters through a network of conducting vessels that, if placed end-to-end, would stretch more than 60,000 miles (more than twice the circumference of the Earth!) ([1](#)). Failure of this remarkable circulatory system begins and ends with cardiac pump failure, which can be categorized as “concerning” (conventional heart failure), “life-threatening” (cardiogenic shock), or “catastrophic” (cardiac arrest).

This chapter focuses on the challenging condition of cardiogenic shock, where management is designed to allow time for the heart to heal from a recent insult (e.g., acute myocardial infarction), or for a definitive procedure to be performed (e.g., bypass surgery). The real challenge in cardiogenic shock, however, is that half of the patients do not survive despite the best management efforts ([2](#)).

PATHOPHYSIOLOGY

Cardiogenic shock can be described as cardiac pump failure that results in inadequate O₂ delivery to tissues and impaired oxidative metabolism. The culmination of this condition is multiorgan failure and a fatal outcome.

Etiologies

Acute myocardial infarction is responsible for about two-thirds of the cases of cardiogenic shock ([3](#)), while the remaining cases can arise from a variety of cardiac disorders (see later).

Acute Myocardial Infarction

Cardiogenic shock develops in 5–10% of cases of acute myocardial infarction ([3](#)), and the mechanisms involved are shown in [Figure 16.1](#). Infarction of the left ventricle is responsible for

about 80% of the cases, and requires loss of at least 40% of the functional mass of the ventricle (2). Infarction of the right ventricle is implicated in less than 5% of cases (3). There are also three infarct-related complications that can produce cardiogenic shock, as described below.

- . Rupture of the papillary muscles (7% of cases) results in acute and life-threatening mitral insufficiency. (*Note:* The papillary muscles attach, via the chordae tendinae, to the leaflets of the mitral valve, and contraction of these muscles prevents prolapse of the mitral valve during ventricular systole.)
- . Rupture of the interventricular septum (4% of cases), is accompanied by a large left-to-right shunt, which floods the pulmonary circulation and diverts flow away from the systemic circulation.
- . Rupture of the ventricular wall (1% of cases) is a catastrophic complication of transmural infarctions, and results in a rapidly-accumulating hemopericardium and pericardial tamponade.

These complications typically appear within 24 hours of the infarction (2), which is contrary to the traditional notion that they are late-onset complications.

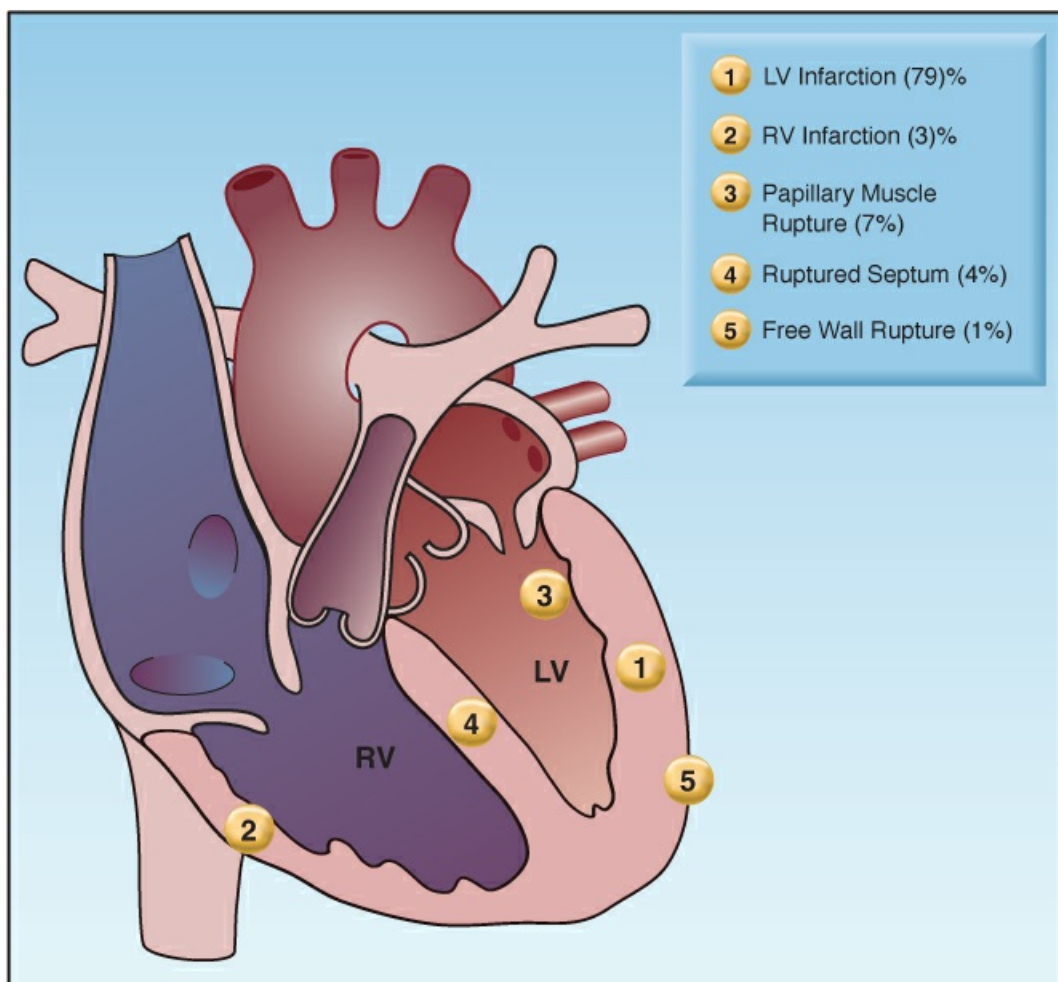


FIGURE 16.1 The mechanisms of cardiogenic shock from acute myocardial infarction. Percentages from Reference 3. **LV** = left ventricle, **RV** = right ventricle.

Other Conditions

The last decade has witnessed an increase in cases of cardiogenic shock that are unrelated to acute myocardial infarction (3), and the major ones are listed in Table 16.1. Acute exacerbations of chronic heart failure have been identified in as many as 30% of cases of cardiogenic shock (2), and stress-induced (Takotsubo’s) cardiomyopathy has emerged as a potential culprit. Other notable conditions are summarized below.

DYNAMIC OUTFLOW OBSTRUCTION: In cases of hypertrophic cardiomyopathy, the flow of blood through the narrowed left ventricular outflow tract can create a suction effect that pulls the mitral valve leaflets anteriorly, and mid-systolic contact of the mitral valve and the interventricular septum creates a “dynamic left ventricular outflow obstruction”(4). The diagnosis of this condition is made with continuous wave Doppler ultrasound in the apical four-chamber view, which is used to calculate the peak systolic pressure gradient across the subaortic region of the left ventricle (5).

Recognition of this condition is important because it is treated differently than other causes of cardiogenic shock: i.e., the goal of management in this case is to slow the heart rate and increase diastolic filling, using nonvasodilating beta-blockers (e.g., metoprolol, nadolol).

CARDIAC SURGERY: Cardiogenic shock is reported in 2–5% of patients who undergo cardiac surgery (6), and this *postcardiotomy shock* can begin in the operating room or in the early postoperative period. Presumed mechanisms include myocardial hibernation or “stunning”, and inadequate cardioprotection. Vasodilation may also play a role in the hypotension (6).

PREGNANCY: There is an idiopathic cardiomyopathy that can appear in the later stages of pregnancy or in the early months following delivery, and is characterized by systolic dysfunction and a variable clinical course. The incidence of this *peripartum cardiomyopathy* in the United States is 1 case per 3,000–4,000 pregnancies (7), and the risk factors include age >30 years, multigestational pregnancy, African heritage, and a family history of cardiomyopathy. This condition can progress to cardiogenic shock and even cardiac arrest, and it has a reported mortality rate of 7–20% in the United States (7).

Obstructive Shock

The category known as “obstructive shock” is essentially cardiogenic shock that is caused by conditions that are extrinsic to the heart. These conditions include acute pulmonary embolism, pericardial tamponade, and tension pneumothorax. The hemodynamic consequences are the same as those of cardiogenic shock, and the management is aimed at correcting the responsible condition.

TABLE 16.1 Causes of Cardiogenic Shock Other than Acute MI	
General	Specific Circumstances
Chronic heart failure with acute decompensation Acute myocarditis Takotsubo’s cardiomyopathy Dynamic outflow obstruction	Postcardiotomy shock Peripartum cardiomyopathy Post-cardiac arrest shock

Hemodynamic Changes

The hemodynamic changes in cardiac pump failure are demonstrated in [Figure 16.2](#). The decrease in cardiac output is accompanied by an increase in cardiac filling pressures (either the central venous pressure or the pulmonary artery wedge pressure, depending on which ventricle has failed), and an increase in systemic vascular resistance (due to activation of both the sympathetic nervous system and the renin-angiotensin-aldosterone system). These changes occur in both heart failure and cardiogenic shock, and *cardiogenic shock is distinguished from heart failure by the presence of tissue hypoperfusion (e.g., hypotension, decreased urine output) and inadequate tissue oxygenation (e.g., elevated plasma lactate levels)*.

Right Heart Failure

Cardiogenic shock from right ventricular (RV) failure has many similarities to shock from left ventricular (LV) failure, but it has a lower mortality rate (8). Some of the features that identify RV failure include an increase in the CVP/PAWP ratio, a dilated RV, and a normal LV ejection fraction. Severe RV dilation can push the interventricular septum into the chamber of the left ventricle, which reduces LV filling and further impairs cardiac stroke output. (The diagnosis and management of RV failure is presented in [Chapter 18](#).)

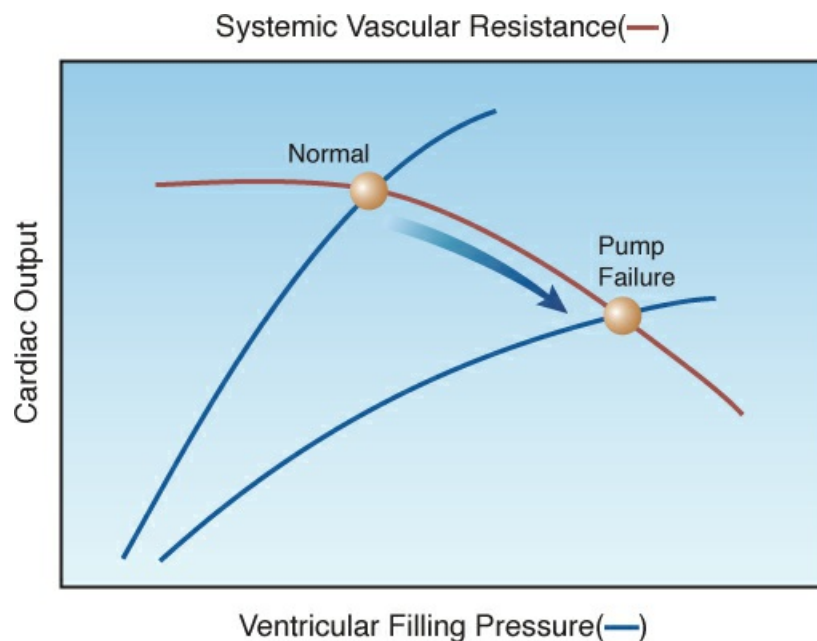


FIGURE 16.2 Graph showing the hemodynamic changes in cardiac pump failure.

Exceptions

The cardiac index (CI) in cardiogenic shock is typically <2.2 L/min/m² (normal CI = 2.4–4.0 L/min/m²) (9), but the cardiac filling pressures are not increased in as many as 30% of patients with infarct-related cardiogenic shock (2). In addition, the systemic vascular resistance may not be elevated when cardiogenic shock is accompanied by systemic inflammation (see later). Finally, hypotension is absent in about 5% of cases of cardiogenic shock (2).

The Microcirculation

Studies using sublingual videomicroscopy have shown that cardiogenic shock is associated with a decrease in capillary perfusion that is independent of the changes in global hemodynamic measures (i.e., cardiac output and blood pressure), and it has a greater impact on mortality than the “macrocirculatory” changes (10). This *microcirculatory dysfunction might explain why correcting the cardiac output and blood pressure in cardiogenic shock does not ensure survival* (10,11). Monitoring microcirculatory perfusion with the PCO₂ gap is potentially useful in this regard (see later).

Systemic Inflammation

Myocardial necrosis triggers a localized inflammatory response (12), which can progress to a systemic inflammatory response. Signs of systemic inflammation (e.g., fever, leukocytosis, etc.) have been reported in 20–40% of patients with cardiogenic shock complicating acute myocardial infarction, and these patients have a lower SVR (due to the vasodilating effects of nitric oxide) and a higher mortality rate (13). This *systemic inflammation has been linked to the microcirculatory dysfunction in cardiogenic shock* (13).

MANAGEMENT

Since most cases of cardiogenic shock are associated with acute myocardial infarction, the management of cardiogenic shock often includes measures aimed at coronary revascularization (i.e., coronary angioplasty, bypass surgery, antithrombotic therapy). These measures are described in [Chapter 20](#); the focus here is on measures that promote tissue perfusion.

Monitoring

The management of cardiogenic shock requires a reliable measure of the cardiac output, and the most reliable measurement available is the thermodilution cardiac output provided by pulmonary artery (PA) catheters (see [Chapter 8](#)). These catheters also provide measurements of the cardiac filling pressures (the central venous and wedge pressures), the systemic vascular resistance, and the oxygen transport variables (see [Table 8.1](#)), which makes them ideal for the management of cardiogenic shock. In fact, the use of PA catheters has been shown to improve outcomes in cardiogenic shock (14), and the most recent guidelines from the American Heart Association emphasizes the value of PA catheters in the management of cardiogenic shock (15).

Tissue Perfusion & Oxygenation

As mentioned earlier, cardiogenic shock can be accompanied by deficits in microcirculatory flow that are independent of the changes in cardiac output, and can persist after the cardiac output and blood pressure are normalized. This highlights the value of monitoring tissue perfusion with the PCO₂ gap (described in [Chapter 14](#)), and with clinical signs of organ perfusion (e.g., urine output).

Tissue oxygenation is probably the most important parameter to monitor in cardiogenic shock, since shock is defined as an abnormality in cellular oxygen utilization (see [Chapter 14](#)). Although it is not possible to directly monitor tissue O₂ levels, the mixed venous O₂ saturation

(SvO₂) can provide information about the balance between O₂ delivery and O₂ consumption, and plasma lactate levels provide an indirect assessment of the adequacy of tissue oxygenation. (These measures are described in [Chapter 9](#).)

Goals of Management

The goals of hemodynamic management in cardiogenic shock are summarized in [Table 16.2](#). Some of these goals are based on the normal range of values for the variable, and some are based on the lowest or highest tolerable levels. An example of the latter instance is presented next.

Optimizing Filling Pressures

As mentioned earlier, the left ventricular filling pressure (i.e., the pulmonary artery wedge pressure) is not always elevated in cardiogenic shock ([2](#)), and when this occurs, volume infusion can be instrumental in augmenting the cardiac output. The target of volume infusion is the highest filling (wedge) pressure that will augment cardiac output without producing pulmonary edema, and this “optimal” pressure is dependent on the colloid osmotic pressure (COP) of plasma. The plasma COP is normally about 28 mm Hg ([16](#)), but a lower COP of 20–25 mm Hg seems more reasonable in critically ill patients, who often have a subnormal plasma albumin concentration (the major determinant of plasma COP). Thus, *the optimal left ventricular filling pressure (i.e., wedge pressure) is considered to be 18–20 mm Hg* ([17](#)).

TABLE 16.2 Goals of Management in Cardiogenic Shock	
Category	Goals
Hemodynamics	PAWP = 19–20 mm Hg CI ≥2.5 L/min/m ² SVRI = 25–30 Wood Units [§] Mean BP ≥65 mm Hg
Tissue Perfusion	PCO ₂ Gap <6 mm Hg Urine output >0.5 mL/kg/h
Tissue Oxygenation	SvO ₂ >50% Plasma lactate <2 mmol/L

[§]Wood Units = mm Hg/L/min/m².

Vasopressor Therapy

Hypotension is almost universal in cardiogenic shock, and vasoconstrictor agents (vasopressors) are used to raise the blood pressure to a systolic BP ≥90 mm Hg or a mean BP ≥65 mm Hg. The vasopressors available for clinical use are described in [Chapter 14](#). *The vasopressor of choice in cardiogenic shock is norepinephrine* ([15,18,19](#)), a catecholamine that produces systemic vasoconstriction and mild cardiac stimulation. One exception to the norepinephrine preference is cardiogenic shock from dynamic LV outflow obstruction, where a pure vasoconstrictor like phenylephrine (which can trigger a reflex bradycardia) is more appropriate ([15](#)).

Norepinephrine

The preference for norepinephrine in cardiogenic shock comes from studies comparing norepinephrine with other popular vasopressors (epinephrine and dopamine), which showed that norepinephrine had a lower risk of tachycardia and troublesome arrhythmias, and was associated with fewer deaths (18,20,21). The mild cardiac stimulation produced by norepinephrine tends to preserve cardiac output in the face of vasoconstriction, at least in the lower dose range.

DOSING: Norepinephrine is given by continuous infusion (without a loading dose) starting at a rate of 5 µg/min (or 0.05 µg/kg/min), and titrating upward every 5 minutes to achieve the desired blood pressure. The usual dose rate in studies of cardiogenic shock is 5–30 µg/min (or 0.05–0.5 µg/kg/min) (20).

Inotropic Therapy

Vasopressor therapy does not alleviate the cardiac pump failure, and can aggravate it. As a result, drugs that increase cardiac contractile force (inotropic agents) are typically added to vasopressor therapy. There are only a few inotropic agents used in this setting, and these are included in Table 16.3, along with their dosing regimens. Inotropic therapy is contraindicated in cardiogenic shock from dynamic LV outflow obstruction.

Dobutamine

The most widely used inotropic agent in cardiogenic shock is dobutamine (18,22), a synthetic catecholamine that stimulates β_1 receptors in the heart, and increases both cardiac stroke output and heart rate. Dobutamine also has mild vasodilator effects, but it does not decrease the blood pressure because the vasodilator effect is offset by the increase in cardiac stroke output. This ability to preserve (and sometimes augment) the blood pressure is the principal advantage of dobutamine over other inotropic agents in cardiogenic shock.

The major disadvantages of dobutamine include an increase in heart rate (which is usually 5–15 beats/min, but can exceed 30 beats/min), and an increase in myocardial O₂ consumption (23). These changes are associated with an increase in cardiac work, which is deleterious in the setting of acute myocardial infarction, and also in the failing myocardium (where cardiac work is already increased).

TABLE 16.3 Inotropic Agents for Cardiogenic Shock

Agent	Dosing Regimen and Comments	
Dobutamine	1. Start infusion at 3–5 µg/kg/min, and increase in increments of 3–5 µg/kg/min, if needed. Usual dose is 3–20 µg/kg/min. 2. Major disadvantage is an increase in heart rate.	
Milrinone	1. Initial dose is 50 µg/kg (over 10 min), followed by an infusion rate of 0.375–0.75 µg/kg/min. Daily dose should not exceed 1.13 mg/kg. 2. Dose adjustments are advised for renal insufficiency: [§]	
	Creatinine Clearance	Infusion Rate
	50 mL/min/1.73 m ² 40	0.43 µg/kg/min 0.38

	30 20	0.33 0.28
	3. Major disadvantage is the risk of hypotension.	
Levosimendan	1. Initial dose is 12 µg/kg (over 10 min), followed by an infusion rate of 0.1 µg/kg/min. After 1 hour, infusion rate can be increased to 0.2 µg/kg/min, if needed. 2. Has a long half-life (80 hrs). 3. Approved for use in about 60 countries, but not in the United States.	

[§]Dosing adjustments are manufacturers recommendation (24).

Milrinone

Milrinone is a phosphodiesterase inhibitor that acts as an inotrope by virtue of inhibiting the breakdown of cyclic AMP, which enhances cyclic AMP-mediated calcium influx into cardiac myocytes. It also acts as a vasodilator, which further promotes cardiac stroke output. However, the vasodilator actions of milrinone can produce troublesome hypotension (23), which is a major drawback in cardiogenic shock. The elimination half-life of milrinone (2.5 hrs) is significantly prolonged in patients with renal insufficiency, and a dosing adjustment is recommended when the creatinine clearance is <50 mL/min/1.73 m² (24), as shown in Table 16.3.

The inotropic effects of milrinone are comparable to those of dobutamine, but there is less risk of troublesome tachycardia (23). Despite this lower risk of tachycardia, milrinone has a significant risk of hypotension, and this limits its popularity as an inotrope in cardiogenic shock. Milrinone is more appropriate for the management of decompensated heart failure without hypotension.

Levosimendan

Levosimendan increases cardiac contractility by sensitizing cardiac myofilaments to calcium, and also promotes vasodilation by facilitating potassium influx into vascular smooth muscle (25). This drug is particularly appealing in infarct-related cardiogenic shock because it dilates coronary arteries and does not stimulate myocardial O₂ consumption. This might explain why studies comparing dobutamine and levosimendan show a mortality benefit with levosimendan (26). However, there is a risk of hypotension with levosimendan, which is a disadvantage. Levosimendan has a long half-life (80 hrs), and infusions are usually limited to 24 hours.

Levosimendan has been a popular inotrope in Europe and South America since it was introduced in 2000 (25), but it is not approved for use in the United States.

Cardiac Workload

The pharmacological management of cardiogenic shock has the unfortunate consequence of increasing the workload of the heart (e.g., vasopressors will increase left ventricular afterload, and inotropes will increase contractility, and can increase heart rate). The increase in cardiac work is counterproductive in the failing myocardium, and seems unlikely to facilitate any improvements in “native” cardiac function. On the other hand, mechanical support devices (described next) are designed to reduce cardiac work, and thus it is important to *avoid delays in proceeding to mechanical cardiac support if there is no evidence of rapid improvement during pharmacological support* (15).

MECHANICAL CARDIAC SUPPORT

Mechanical cardiac support (MCS) is generally reserved for cases of refractory cardiogenic shock where improvement in cardiac function is an expectation (usually after myocardial infarction or cardiac surgery), or another intervention is planned (e.g., coronary bypass surgery) (27).

Intra-Aortic Balloon Pump

The intra-aortic balloon pump (IABP) is the oldest form of MCS (introduced in the 1960s), and continues to be used despite evidence that it does not improve survival (28). However, *studies comparing the IABP with other forms of MCS have shown no improvement in survival with any of the methods* (15). (This seems to be a testament to the whole MCS approach to cardiogenic shock, however there are individual patients who benefit from MCS, and it is often difficult to predict who will benefit prior to the intervention.) The IABP is the most readily available form of MCS in hospitals other than large, tertiary medical centers, but it is only a temporary measure. It is contraindicated in patients with aortic valve insufficiency or aortic dissection.

The Method

The intra-aortic balloon is an elongated polyurethane balloon that is inserted percutaneously into the femoral artery and advanced up the aorta until the tip lies just below the origin of the left subclavian artery (see [Figure 16.3](#)). A pump attached to the balloon uses helium, a low density gas, to rapidly inflate and deflate the balloon (inflation volume is generally 35 to 40 mL). Inflation begins at the onset of diastole, just after the aortic valve closes (the R wave on the ECG is a common trigger). The balloon is then deflated at the onset of ventricular systole, just before the aortic valve opens (during isovolumic contraction). This pattern of balloon inflation and deflation produces two changes in the aortic pressure waveform, which are illustrated in [Figure 16.3](#).

- . Inflation of the balloon during diastole increases the peak diastolic pressure, which augments coronary blood flow (which occurs predominantly during diastole), and also increases the mean arterial pressure (equivalent to the integrated pressure under the aortic pressure waveform), which augments systemic blood flow.
- . Deflation of the balloon creates a suction effect that reduces pressure in the aorta when the aortic valve opens, and this decreases left ventricular afterload and augments ventricular stroke output.

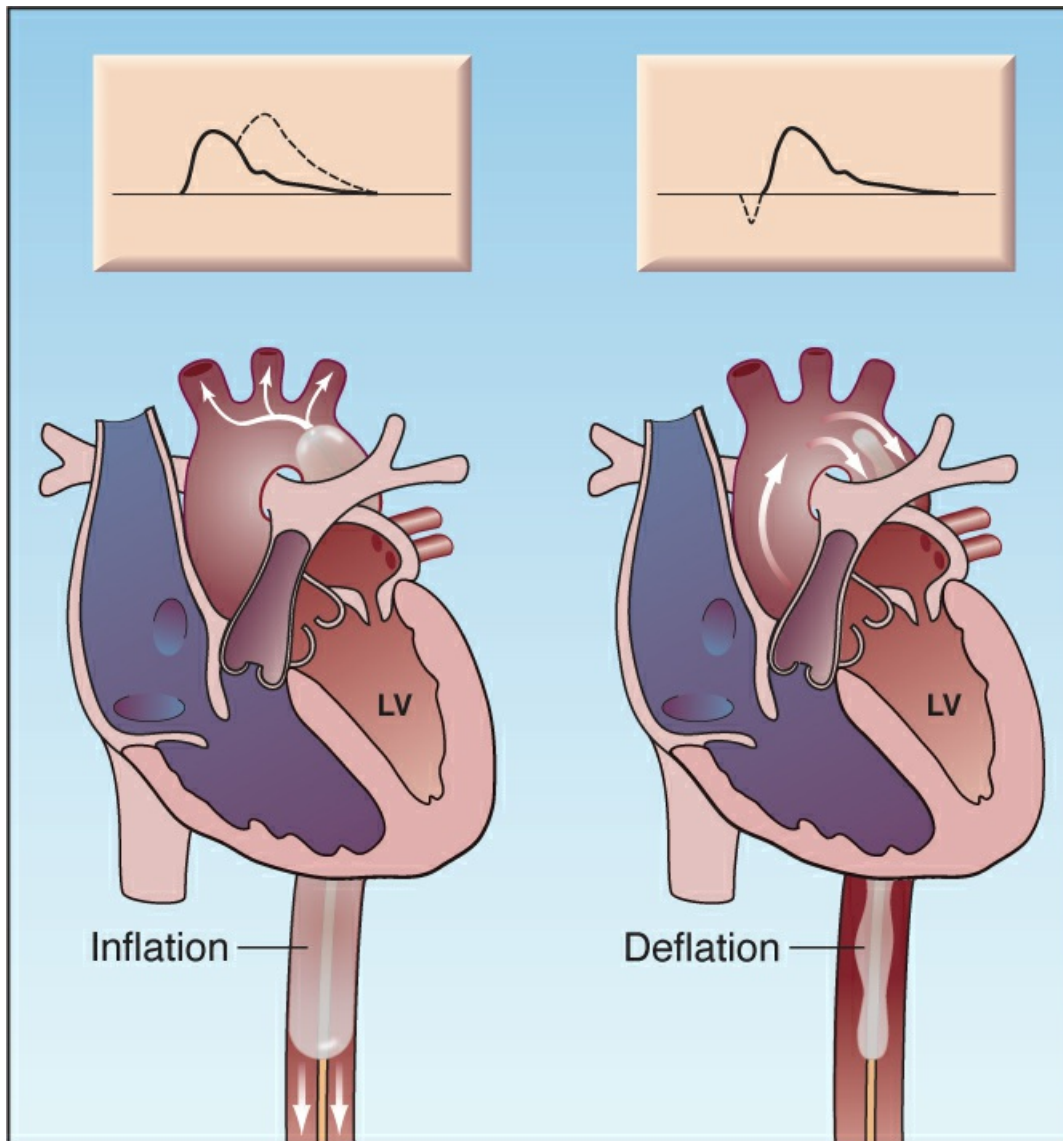


FIGURE 16.3 Intra-aortic balloon counterpulsation, showing balloon inflation in early diastole (on the left), and balloon deflation in late diastole (on the right). The associated changes in the aortic pressure (indicated by the dotted lines) are shown above each maneuver. The arrows show the direction of enhanced flow.

The IABP thus reduces cardiac work by unloading the left ventricle while also increasing coronary blood flow and systemic flow.

Complications

The principal concern with IABP is limb ischemia, which can appear while the balloon is in place or shortly after balloon removal. Most cases are the result of in-situ thrombosis at the catheter insertion site, but aortoiliac injury may also be responsible. Loss of distal pulses alone does not warrant removal of the balloon as long as sensorimotor function in the legs is intact (29). Loss of sensorimotor function in the legs should always prompt immediate removal of the device, and surgical intervention may be required.

Other complications of IABP include septicemia, balloon rupture, peripheral neuropathy, and pseudoaneurysm. Fever is reported in 50% of patients during IABP support, but bacteremia is reported in only 15% of patients (30).

Impella Catheters

Impella® catheters (ABIOMED, Inc., Danvers, MA) provide flow assistance for ventricular output. These catheters are primarily used as left-ventricular assist devices (LVADs), but there is also a catheter designed to assist the right ventricle (an RVAD).

An example of an Impella catheter shown in Figure 16.4. The catheter is equipped with a miniature centrifugal pump (i.e., the type of pump that uses rotational “propeller” blades to create non-pulsatile flow), and it is inserted percutaneously into a femoral artery and advanced into the left ventricle. (The pigtail at the tip of the catheter allows the catheter to be advanced safely.) The pump moves blood from the left ventricle to the proximal aorta at flow rates of 2.5–5.5 L/min (flow rates differ with different catheters). The device shown in Figure 16.4 monitors the native cardiac output, and automatically adjusts the pump flow to achieve a (pre-selected) total cardiac output (31).

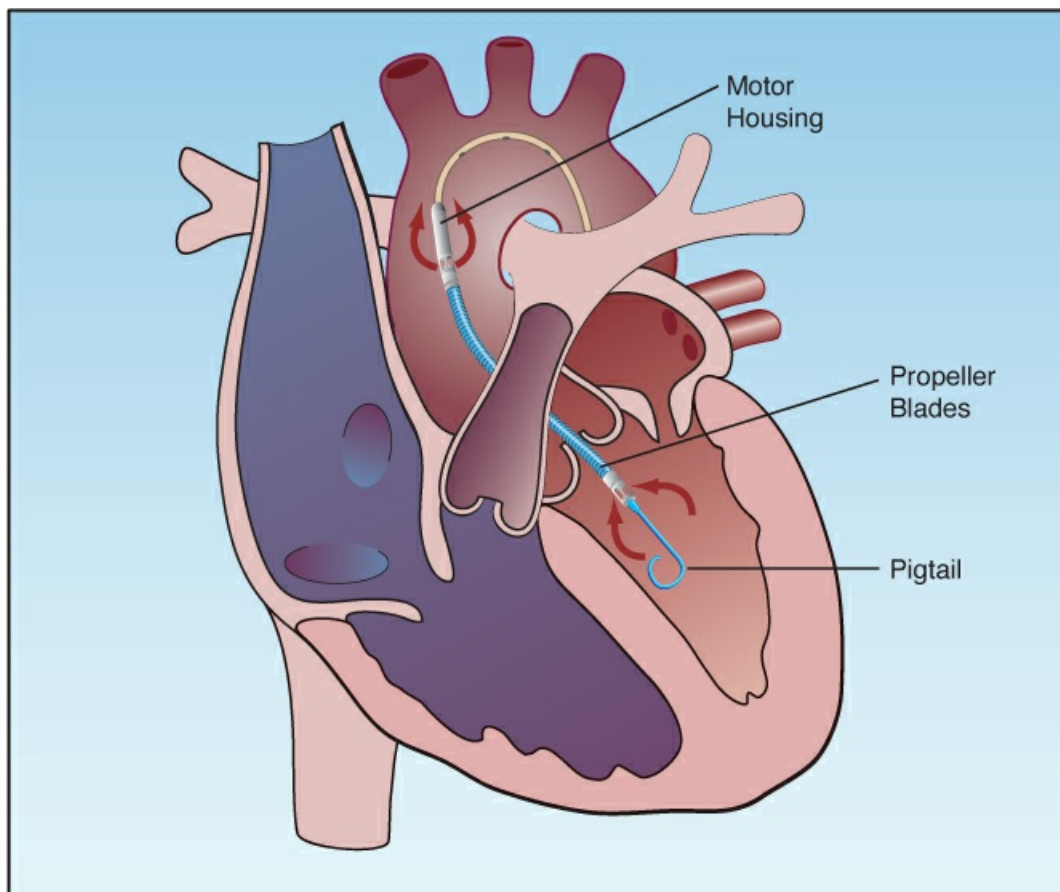


FIGURE 16.4 An Impella catheter, which is equipped with a miniature centrifugal pump. The tip of the catheter is placed in the left ventricle so that the pump’s rotational propellers are positioned across the aortic valve. The pump then transfers blood from the ventricle to the proximal aorta (see directional arrows) to unload the ventricle. See text for further explanation.

The hemodynamic effects of Impella catheters are similar to the IABP; i.e., they unload the left ventricle and increase cardiac output, while also increasing coronary blood flow (32). This similarity also extends to survival value, since Impella catheters have not shown a survival advantage over IABP in patients with infarct-related cardiogenic shock (28,33). Impella catheters have more of an advantage in patients with end-stage left ventricular failure, since they can be implanted for long-term use. These catheters are contraindicated in patients with aortic valve disease or a prosthetic aortic valve, and in patients with an LV thrombus.

Complications

The complications of the Impella catheter are similar in type to the IABP, but there is evidence that major bleeding and limb ischemia are more common with Impella catheters (33,34). The incidence of significant hemolysis (5–10%) is also greater with Impella catheters than with other forms of mechanical support (35).

ECMO

Extracorporeal membrane oxygenation (ECMO) is designed to support both circulatory flow and pulmonary gas exchange. Veno-arterial ECMO (VA-ECMO) is used for cardiogenic shock, and is gaining in popularity for the management of refractory cardiogenic shock associated with acute myocardial infarction and cardiac surgery. VA-ECMO offers some advantages, including the ability to provide full biventricular support (matched only by combining RV and LV Impella catheters), and the ability to support oxygenation and CO₂ removal, which is advantageous in patients with respiratory failure (a common complication of cardiogenic shock). *VA-ECMO is the preferred method of mechanical support in cases of cardiogenic shock associated with respiratory failure* (36).

The basic elements of a VA-ECMO circuit are shown in Figure 16.5. Vascular access can be achieved by percutaneous cannulation of the femoral artery and vein. The femoral vein catheter is advanced along the inferior vena cava until the tip is situated close to the right atrium. Blood from the femoral vein catheter is passed through a centrifugal pump (which generates the desired pressures and flow rates) and then flows through a membrane oxygenator before being warmed to 37° C and returned via the femoral artery catheter. Retrograde flow up the aorta is intended to reach the aortic arch so it can enhance cerebral blood flow. Flow rates vary according to each patient's native cardiac output, but they can reach 5–6 L/min to provide full circulatory support. As in the other forms of mechanical support, full anticoagulation is necessary.

Complications

The complications of VA-ECMO are varied and considerable. Reported complications and their frequency of appearance are as follows (36): major bleeding (41%), sepsis (30%), limb ischemia (17%), lower extremity compartment syndrome (10%), stroke (6%), and lower limb amputation (5%).

LV AFTERLOAD: VA-ECMO has one problem that is not shared by the other forms of mechanical support: i.e., the retrograde flow in the aorta increases left ventricular (LV) afterload, and this can be severe enough to reduce LV output and cause acute pulmonary edema (36). Efforts to tackle this problem have included combining IABP or Impella catheters with VA-ECMO to

reduce LV afterload, and draining blood from the left ventricle to reduce end-diastolic pressure (known as “venting”) (37).

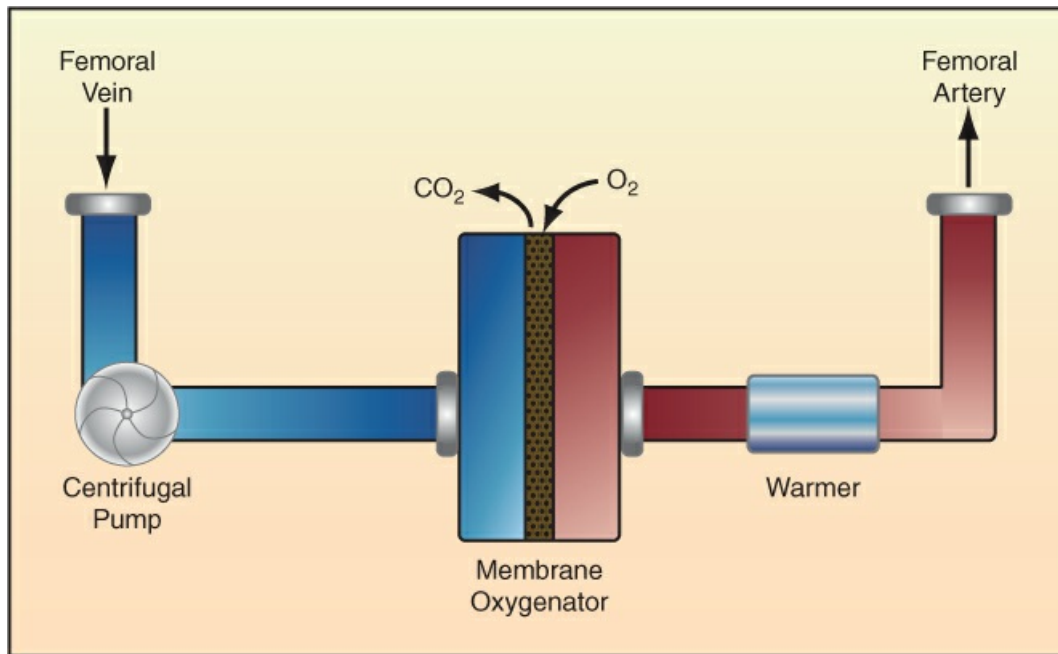


FIGURE 16.5 Schematic design of a VA-ECMO circuit. See text for explanation.

Outcomes

The largest registry of ECMO patients has reported a 42% survival rate to hospital discharge when VA-ECMO is used for cardiogenic shock (38). A comparison of survival rates with VA-ECMO versus other forms of mechanical support (IABP or LVAD) has had inconsistent results; i.e., survival rates have been better with VA-ECMO in some reports (39) and better with the Impella catheter in other reports (40).

Device Selection

The choice of mechanical support device will largely be determined by availability and individual preferences at your hospital. VA-ECMO is the method of choice for cardiogenic shock associated with respiratory failure, and there is a trend in favor of VA-ECMO for all mechanical support. However, ECMO is not available in many hospitals, and mechanical support with IABP or an LVAD may be the only options (patients can then be transferred for ECMO if there is no improvement). Impella catheters are preferred to IABP in many hospitals, even though the outcomes appear to be the same with both devices.

A FINAL WORD

The following information in this chapter deserves emphasis.

- . The management of cardiogenic shock is aimed at “buying time” until either the heart begins to heal from an acute insult (e.g., an acute coronary event), or another intervention is

performed (e.g., bypass surgery).

- . Pharmacological support is an immediate need for maintaining tissue perfusion, but this approach increases the workload of the heart, and is counterproductive. For this reason, proceeding to mechanical cardiac support should be a priority if there is no evidence of improvement with pharmacological support.
- . Mechanical support devices can “unload” the ventricle (intra-aortic balloon pump), provide flow assistance for ventricular output (Impella catheters), or provide support for gas exchange and circulatory blood flow (VA-ECMO).
- . VA-ECMO is the support method of choice if cardiogenic shock is accompanied by respiratory failure. Otherwise, there is no consistent evidence showing that one method is superior to the others for a satisfactory outcome.
- . Avoiding delays to initiation of mechanical support will optimize outcomes. Mechanical support that is initiated after the onset of multiorgan failure has been described as a “bridge to nowhere” (36).

References

- a. Schrödinger E. What is Life? New York: Cambridge University Press, 1944.
1. Vogel S. Vital Circuits. New York: Oxford University Press, 1992;1–17.

Pathophysiology

2. van Diepen S, Katz JN, Albert NM, et al; for the American Heart Association Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing; Council on Quality of Care and Outcomes Research; and Mission: Lifeline. Contemporary management of cardiogenic shock: A scientific statement from the American Heart Association. *Circulation* 2017; 136:e232–e268.
3. Berg DD, Bohula EA, Morrow DA. Epidemiology and causes of cardiogenic shock. *Curr Opin Crit Care* 2021; 27:401–408.
4. Sobozyk D. Dynamic left ventricular outflow obstruction: underestimated cause of hypotension and hemodynamic instability. *J Ultrasonography* 2014; 4:421–427.
5. Panza JA, Petrone RK, Fananapazir L, Maron BJ. Utility of continuous wave Doppler echocardiography in the noninvasive assessment of left ventricular outflow tract pressure gradient in patients with hypertrophic cardiomyopathy. *J Am Coll Cardiol* 1992; 19:91–96.
6. Masud F, Gheewala G, Giesecke M, et al. Cardiogenic shock in perioperative and intraoperative settings: A team approach. *Methodist DeBakey Cardiovasc J* 2020; 16:e1–e7.
7. Davis MB, Arany Z, McNamara DM, et al. Peripartum cardiomyopathy. JACC state-of-the-art review. *J Am Coll Cardiol* 2020; 75:207–221.
8. Josiassen J, Helgestad OKL, Møller JE, et al. Cardiogenic shock due to predominantly right ventricular failure complicating acute myocardial infarction. *Eur Heart J: Acute Cardiovasc Care* 2021; 10:33–39.
9. Menon V, White H, LeJemtel T, et al. The clinical profile of patients with suspected cardiogenic shock due to predominant left ventricular failure: a report from the SHOCK Trial Registry: Should we emergently revascularize Occluded Coronaries in cardiogenic shock? *J Am Coll Cardiol* 2000; 36(suppl A):1071–1076.
10. Wijntjens GWM, Fengler K, Feurnau G, et al. Prognostic implications of microcirculatory perfusion versus macrocirculatory perfusion in cardiogenic shock: a CULPRIT-SHOCK substudy. *Eur Heart J: Acute Cardiovasc Care* 2020; 9:108–119.
11. Jung C, Feurnau G, de Waha S, et al. Intraaortic balloon counterpulsation and microcirculation in cardiogenic shock complicating myocardial infarction: an IABP-SHOCK II substudy. *Clin Res Cardiol* 2015; 104:679–687.
12. Frangogiannis NG, Smith CW, Entman ML. The inflammatory response in myocardial infarction. *Cardiovasc Res* 2002; 53:31–47.
13. Bertini P, Guarracino F. Pathophysiology of cardiogenic shock. *Curr Opin Crit Care* 2021; 27:409–415.

Management

14. Bertaina M, Galluzzo A, Rossello X, et al. Prognostic implications of pulmonary artery catheter monitoring in patients with cardiogenic shock: A systematic review and meta-analysis of observational studies. *J Crit Care* 2022; 69:154024.
15. Henry TD, Tomey MI, Tamis-Holland JE, et al; on behalf of the American Heart Association Interventional Cardiovascular Care Committee of the Council on Clinical Cardiology; Council on Arteriosclerosis, Thrombosis, and Vascular Biology; and Council on Cardiovascular and Stroke Nursing. Invasive management of acute myocardial infarction complicated by

cardiogenic shock. A scientific statement from the American Heart Association. *Circulation* 2021; 143:e815–e829.

16. Guyton AC, Hall JE. Textbook of Medical Physiology. 10th ed., Philadelphia: W.B. Saunders, Co., 2000;169–170.
17. Franciosa JA. Optimal left heart filling pressure during nitroprusside infusion for congestive heart failure. *Am J Med* 1983; 74:457–464.
18. DeBacker D, Ortiz JA, Levy B. The medical treatment of cardiogenic shock: cardiovascular drugs. *Curr Opin Crit Care* 2021; 27:426–432.
19. Thiele H, Ohman EM, de Waha-Thiele S, et al. Management of cardiogenic shock complicating myocardial infarction: an update 2019. *Eur Heart J* 2019; 40:2671–2683.
20. Rui Q, Jiang Y, Chen M, et al. Dopamine versus norepinephrine in the treatment of cardiogenic shock: a PRISMA-compliant meta-analysis. *Medicine* 2017; 96:e8402.
21. Levy B, Clere-Jehl R, Legras A, et al. Epinephrine versus norepinephrine for cardiogenic shock after acute myocardial infarction. *J Am Coll Cardiol* 2018; 72:173–182.
22. Desjardin JT, Teerlink JR. Inotropic therapies in heart failure and cardiogenic shock: an educational review. *Eur Heart J: Acute Cardiovasc Care* 2021; 10:676–686.
23. Bayram M, De Luca L, Massie B, Gheorghiade M. Reassessment of dobutamine, dopamine, and milrinone in the management of acute heart failure syndromes. *Am J Cardiol* 2005; 96(Suppl): 47G–58G.
24. Milrinone Lactate Injection (package insert). Hospira, Inc, Lake Forest, IL, 2021.
25. Papp Z, Agostoni P, Alvarez J, et al. Levosimendan efficacy and safety: 20 years of SIMDAX in clinical use. *Cardiac Failure Rev* 2020; 6:e19.
26. Schumann J, Henrich EC, Strobl H, et al. Inotropic agents and vasodilator strategies for cardiogenic shock or low cardiac output syndrome. *Cochrane Database Syst Rev* 2018; 1:CD009669.

Mechanical Support

27. Rob D, Bêlohlávek J. The mechanical support of cardiogenic shock. *Curr Opin Crit Care* 2021; 27:440–446.
28. Thiele H, Zeymer U, Neumann FJ, et al; IABP-SHOCK II Trial Investigators. Intraaortic balloon support for myocardial infarction with cardiogenic shock. *N Engl J Med* 2012; 367:1287–1296.
29. Baldyga AP. Complications of intra-aortic balloon pump therapy. In Maccioli GA, ed. *Intra-aortic balloon pump therapy*. Philadelphia: Williams & Wilkins, 1997;127–162.
30. Crystal E, Borer A, Gilad J, et al. Incidence and clinical significance of bacteremia and sepsis among cardiac patients treated with intra-aortic balloon counterpulsation pump. *Am J Cardiol* 2000; 86: 1281–1284.
31. Impella 5.5® with SmartAssist® Instructional Manual. ABIOMED, Inc., 2022.
32. Remmelink M, Sjauw KD, Henriques JP, et al. Effects of left ventricular unloading by Impella recover LP2.5 on coronary hemodynamics. *Catheter Cardiovasc Interv* 2007; 70:532–537.
33. Schrage B, Ibrahim K, Loehn T, et al. Impella support for acute myocardial infarction complicated by cardiogenic shock. *Circulation* 2019; 139:1249–1258.
34. Dhruva SS, Ross JS, Mortazavi BJ, et al. Association of use of an intravascular microaxial left ventricular assist device vs intra-aortic balloon pump with in-hospital mortality and major bleeding among patients with acute myocardial infarction complicated by cardiogenic shock. *JAMA* 2020; 323:734–741.
35. Miller PE, Solomon MA, McAreavey D. Advanced percutaneous mechanical circulatory support devices for cardiogenic shock. *Crit Care Med* 2017; 45:1922–1929.
36. Combes A, Price S, Slutsky AS, et al. Temporary circulatory support for cardiogenic shock. *Lancet* 2020; 396:199–212.
37. Cevasco M, Takayama H, Ando M, et al. Left ventricular distension and venting strategies for patients on venoarterial extracorporeal membrane oxygenation. *J Thorac Dis* 2019; 11:1676–1683.
38. Thiagarajan RR, Barbaro RP, Rycus PT, et al. Extracorporeal Life Support Organization registry international report 2016. *ASAIO J* 2017; 63: 60–67.
39. Ouweneel DM, Schotborgh JV, Limpens J, et al. Extracorporeal life support during cardiac arrest and cardiogenic shock: a systematic review and meta-analysis. *Intensive Care Med* 2016; 42:1922–34.
40. Abusnina W, Ismayl M, Al-abdouh A, et al. Impella versus extracorporeal membrane oxygenation in cardiogenic shock: A systematic review and meta-analysis. *Shock* 2022; 58:349–357.

Inflammatory Shock Syndromes

Inflammation is not itself considered a disease but a salutary operation... but when it cannot accomplish that salutary purpose... it does mischief.

John Hunter, MD ([a](#))

The introductory quote is from an 18th-Century British surgeon with an insatiable curiosity, who was known for his frequent grave-robbing soirees (for research material), and for a reckless self-experiment in which he inoculated himself with the purulent discharge from a patient with venereal disease, and subsequently developed both gonorrhea and syphilis ([1](#)). On a more positive note, John Hunter wrote the first definitive work on inflammation, where he noted that inflammation has two opposing “faces” (like a Janus face): i.e. it normally acts to promote healing from tissue injury or infection, but it can also become a *source* of tissue injury ([2](#)). Fast-forward over two hundred years, and the destructive face of inflammation has emerged as a major factor in a multitude of diseases, from acute (e.g., asthma) to chronic (e.g., atherosclerosis) to age-related (e.g., Alzheimer’s) conditions.

This chapter describes three life-threatening conditions where inflammation plays a major role; i.e., septic shock, toxic shock, and anaphylactic shock. These conditions are examples of what happens when inflammation “does mischief”.

INFLAMMATORY INJURY

The inflammatory response is a complex affair that is designed to eliminate external threats (e.g., from invading microbes) and to “cleanse” and heal damaged tissues. This operation is not typically harmful to the host, however severe or persistent inflammation can produce widespread tissue injury. Inflammatory tissue injury is especially problematic because it can become self-sustaining; i.e., the inflammatory injury can trigger another inflammatory response, which will produce more tissue injury, and so on. This condition of progressive inflammatory injury is known as *malignant inflammation*, and it plays a major role in the multiorgan failure that can complicate septic shock (see later) ([3,4](#)).

Neutrophil Activation

In the initial stage of the inflammatory response, circulating neutrophils experience a precipitous rise in cellular O_2 consumption (up to 20-fold) that lasts for 15–20 minutes (5). This is known as the *respiratory burst*, which is a misnomer, because the surge in O_2 consumption is not dedicated to the production of high-energy ATP molecules (i.e., cell respiration). Instead, the respiratory burst generates noxious chemical derivatives of oxygen that are used to kill invading microbes (5).

The reactions involved in the respiratory burst are shown in [Figure 17.1](#). The trigger for this reaction sequence is activation of a specialized enzyme (NADPH oxidase) on the cell membrane, which initiates a series of reactions that adds electrons (one at a time) to oxygen. This generates a series of highly-reactive intermediates that include the *superoxide radical*, *hydrogen peroxide* and the *hydroxyl radical*. (A radical is an atom or molecule with an unpaired electron.) Neutrophils also have a myeloperoxidase enzyme that converts hydrogen peroxide to *hypochlorite*, a powerful germicidal agent that is the active ingredient in household bleach. These *reactive oxygen species* (ROS) are stored in cytoplasmic granules, and are released during neutrophil degranulation.

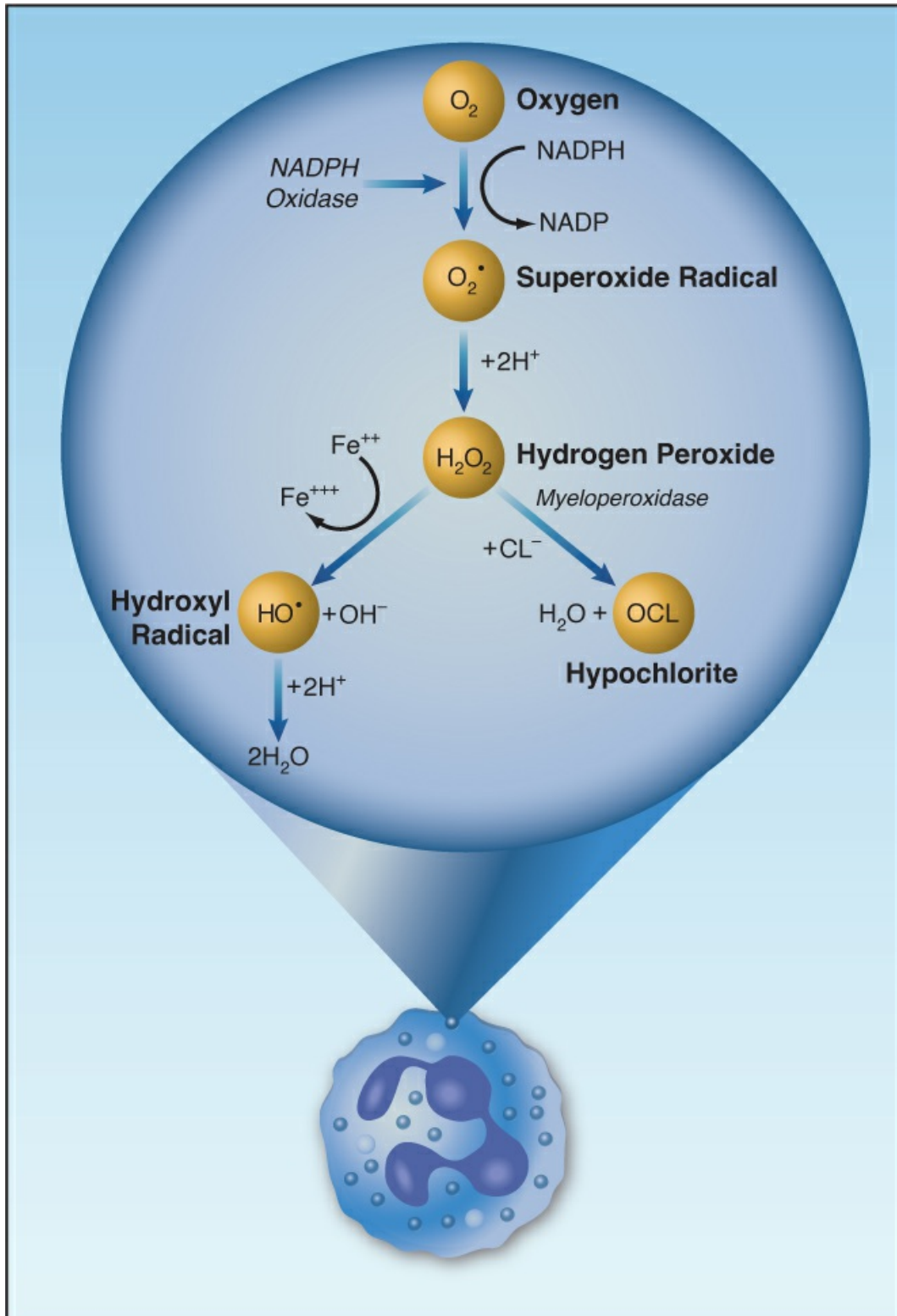


FIGURE 17.1 The sequence of reactions involved in the activation of neutrophils, which generates a series of reactive oxygen species (ROS) that are used to kill invading microbes. Free radicals are indicated by a superscripted dot. See text for further explanation.

The ROS are powerful oxidizing agents that are lethal for invading microbes. The role of

ROS in eradicating infectious organisms is demonstrated by a condition known as *chronic granulomatous disease*. This is a rare genetic disorder (with an incidence of 1:200,000 births in U.S.) in which the NADPH oxidase enzyme in neutrophils is defective, and patients with this disorder suffer from recurrent and life-threatening bacterial and fungal infections (6).

Oxidant Injury

ROS are also capable of damaging the tissues of the host, including the extracellular matrix, parenchymal cells, and the endothelium and its protective glycocalyx (7–9). The oxidizing effects of ROS can damage all vital cell components, including membrane lipids, all types of cellular proteins, and DNA molecules. This *oxidant injury* occurs when ROS activity exceeds the protective actions of endogenous antioxidants; a condition known as *oxidant stress*.

(Note: The destructive actions of oxygen on organic matter is evident when you consider why food is stored in vacuum-sealed containers, and why you wrap your sandwiches in cellophane.)

Inflammatory tissues are flooded with ROS, and the superoxide radical has been implicated as the source of inflammatory pain (10). Oxidant tissue injury usually begins to appear when the inflammation is persistent or progressive (possibly due to depletion of protective antioxidants). Furthermore, hydrogen peroxide moves freely through tissues, and can enter the bloodstream and reach distant organs, where it can enter cells and generate hydroxyl radicals when it encounters iron in the reduced state (Fe^{2+}) (see Figure 17.1). This is one possible mechanism for the multiorgan failure that can develop in patients with inflammatory lung injury (see Chapter 24), or sepsis (see later), as well as the postresuscitation syndrome after a cardiac arrest (see Chapter 21), which is a type of disseminated reperfusion injury.

Endothelial Dysfunction

Like neutrophils, endothelial cells are equipped with a NADPH oxidase enzyme that is activated in the early stages of the inflammatory response, and the resulting production of ROS is instrumental in promoting leukocyte adhesion to the endothelium and the migration of leukocytes into the tissues (11). However, inflammation also promotes endothelial dysfunction from oxidant stress (12), and this leads to increased capillary permeability and edema formation.

Oxidant stress at the endothelial level also reduces nitric oxide levels via the following reaction (13):



where $\text{O}_2\cdot$ is the superoxide radical, $\text{NO}\cdot$ is nitric oxide, and ONOO^- is peroxynitrite. This reaction not only eliminates the beneficial actions of nitric oxide in promoting circulatory blood flow, it also generates a potent oxidative toxin (peroxynitrite) capable of extensive tissue injury (13).

Inflammatory Organ Injury

Inflammation can damage any of the vital organs, but the organs most often affected are listed in Table 17.1. These conditions pertain only to the inflammatory injury that occurs in critically ill patients, and they do not include the multitude of diseases caused by chronic inflammatory injury (e.g., rheumatoid arthritis, chronic obstructive lung disease). The most notable example of

inflammatory organ injury in critically ill patients is the acute respiratory distress syndrome (ARDS), which is a leading cause of acute respiratory failure that requires mechanical ventilation. (ARDS is the focus of [Chapter 24](#).)

TABLE 17.1 Inflammatory Organ Injury in Critically Ill Patients	
Organ System	Clinical Condition
Brain	Septic encephalopathy
Bone Marrow	Anemia of critical illness
Circulatory System	Septic shock, Anaphylactic shock
Kidneys	Acute Kidney Injury
Lungs	Acute respiratory distress syndrome
Peripheral Nerves	Critical illness polyneuropathy
Skeletal Muscle	Critical illness myopathy, Rhabdomyolysis

The morbidity and mortality associated with inflammatory organ injury is not only related to the severity of involvement in individual organs, but also to the number of organs involved. This latter relationship is shown in [Figure 17.2](#), which includes surveys from the United States (14) and Europe (15) showing a direct relationship between the mortality rate and the number of organ failures related to inflammation.

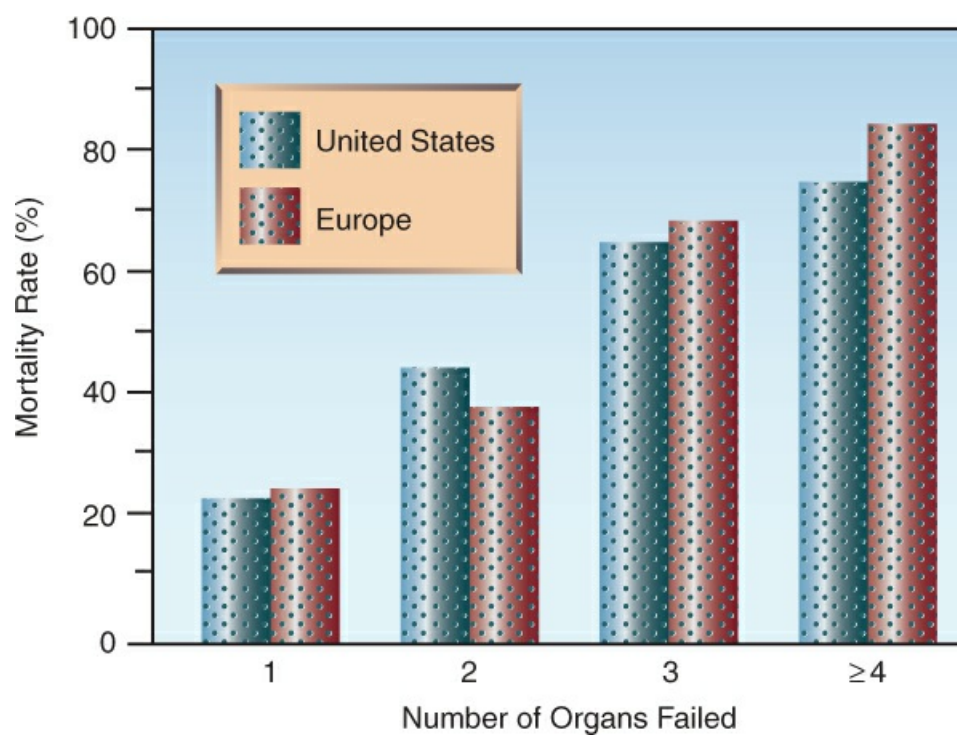


FIGURE 17.2 The relationship between mortality rate and the number of inflammation-related organ failures. Data from References 12 and 13.

SEPTIC SHOCK

Sepsis is responsible for one of every 2 to 3 in-hospital deaths in the United States (16), and one of every 5 deaths worldwide (17). The mortality rate varies by age and geographic locale: for septic shock in the United State, the mortality rate is about 30% when it is present on admission, and about 50% when it develops after admission (18). Despite the presumption of sepsis as infection, *the mortality rate in sepsis is not related to the causative organism, including multidrug-resistant organisms* (19). This observation is consistent with the notion that the major problem in sepsis is not the infection, but the host response to infection (see next).

Definitions

The current definitions of sepsis and septic shock are summarized below (20).

- . Sepsis is an infection (suspected or confirmed) that is associated with life-threatening organ dysfunction, and is the result of a dysregulated host response to infection.
- . Septic shock is a subset of sepsis that is characterized by the need for vasopressor support, and by a plasma lactate >2 mmol/L.

These definitions were introduced in 2016, and they replaced the prior definitions established in 1992 (21), which defined sepsis as an infection that was accompanied by a systemic inflammatory response (i.e., fever, leukocytosis, etc.) (21), with the latter condition being elevated to a syndrome; i.e., the “systemic inflammatory response syndrome”, or SIRS. However, SIRS is a nonspecific finding in critically ill patients, as fewer than 50% of patients with SIRS have a documented infection (22,23), so SIRS was abandoned as a diagnostic criterion for sepsis. However, the definition of sepsis continues to be problematic, as explained next.

A Better Definition

The presence of infection is a central feature in the definition of sepsis and septic shock. However, there is often no documented infection in cases of sepsis; e.g., *in a survey of 2.5 million patients with sepsis and septic shock, there was no documented infection in 70% of the cases* (18). Since the clinical suspicion of infection in cases of sepsis is typically based on the presence of a systemic inflammatory response (fever, leukocytosis, etc.), it seems likely that *sepsis is a dysregulated host response to inflammation, not infection*.

Pathophysiology

Regardless of whether the sepsis is a dysregulated response to infection or inflammation, septic shock is distinguished from sepsis by the presence of hemodynamic instability and mitochondrial dysfunction.

Hemodynamic Alterations

The hemodynamic aberrations in septic shock are summarized below:

- . The principal hemodynamic problem is hypotension from systemic vasodilation. This has been attributed to the enhanced production of nitric oxide by the vascular endothelium (24), but the inactivation of nitric oxide by oxidant stress (see Equation 17.1) suggests the involvement of

other factors.

- . Oxidant injury in the vascular endothelium leads to fluid extravasation and hypovolemia, which aggravates the hypotension.
- . Proinflammatory cytokines promote cardiac dysfunction (both systolic and diastolic dysfunction), but the cardiac output is usually increased as a result of tachycardia and a decrease in left ventricular afterload (from vasodilation) (25). The cardiac output begins to decline in the advanced stages of shock, which carries a poor prognosis.
- . Despite the increased cardiac output, splanchnic blood flow is typically reduced in septic shock (24). This can lead to disruption of the bowel mucosa and translocation of enteric pathogens into the systemic circulation, which aggravates the problem.

Because of the high cardiac output and peripheral vasodilation, septic shock is also known as *hyperdynamic shock* or *warm shock*.

Mitochondrial Dysfunction

The organ dysfunction in sepsis and septic shock is attributed to a defect in the production of high-energy ATP molecules in mitochondria (26,27). This condition is attributed to oxidative damage in the electron transport chain (28), and to cytokine-mediated inhibition of pyruvate dehydrogenase, the enzyme that facilitates the entry of pyruvate into mitochondria (see Figure 9.5) (29). The resulting decrease in O₂ usage in mitochondria would explain the observation shown in Figure 17.3, where the PO₂ in skeletal muscle is *increased* in patients with severe sepsis (30).

The decrease in mitochondrial O₂ consumption in sepsis is not consistent with the observed increase in whole-body O₂ consumption in patients with sepsis. This discrepancy can be resolved by proposing that the increased O₂ consumption in sepsis is a reflection of the marked increase in O₂ consumption that occurs during neutrophil activation (31).

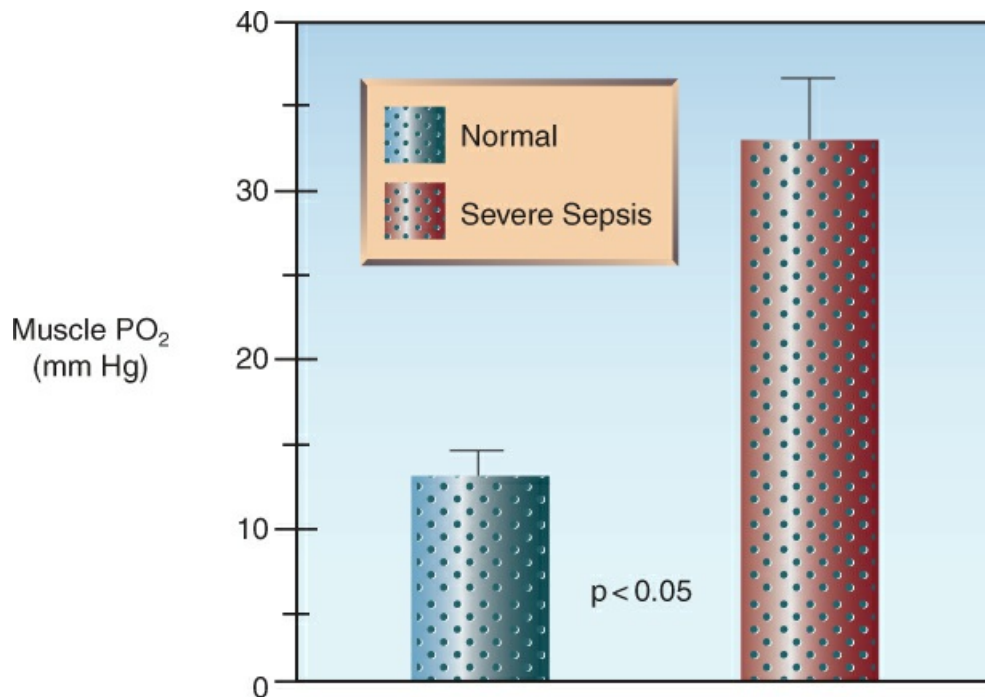


FIGURE 17.3 Direct measurements of tissue PO₂ in the forearm muscles of healthy volunteers and patients with severe sepsis. The height of the columns represents the mean value for each group, and the crossbars represent the standard error of the mean. Data from Reference 30.

IMPLICATIONS: The mitochondrial dysfunction in sepsis and septic shock has two very important implications:

- . Since tissue O₂ levels are not impaired in septic shock, the liberal use of supplemental O₂ to promote tissue oxygenation is not justified, and could be harmful (by promoting oxidant stress).
- . The increase in plasma lactate levels in septic shock is not the result of inadequate tissue oxygenation. Instead, the emerging paradigm is that lactate production increases during periods of metabolic stress, where lactate serves as an alternative energy source (see [Table 9.6](#)).

Monitoring

The approach to monitoring in patients with clinical shock is described in [Chapter 14](#), but the following information about septic shock deserves mention.

- . The decrease in mitochondrial O₂ usage in septic shock is associated with a decrease in O₂ extraction from the microcirculation (i.e., a decrease in the arteriovenous O₂ saturation difference) As a result, the central venous O₂ saturation (ScvO₂) will be inappropriately increased, and will be misleading as a measure of the adequacy of O₂ delivery. (Oxygen extraction is described in [Chapter 9](#).)
- . The PCO₂ gap is not influenced by systemic oxygenation, and can be useful for identifying tissue hypoperfusion (see [Chapter 14](#)).
- . Although lactate is not a marker of anaerobic metabolism in septic shock, plasma lactate levels

have prognostic significance (see [Figure 9.4](#)), and normalization of lactate levels is an important goal of management.

Management

The initial management of septic shock is outlined in [Table 17.2](#), using recommendations from the most recent clinical practice guidelines ([32](#)).

Volume Resuscitation

Volume infusion should be the initial intervention in any patient with a shock syndrome. The recommended fluid regimen for septic shock is 30 mL/kg (ideal body weight) of a crystalloid fluid, infused within the first 3 hours of diagnosis ([32](#)). The need for repeat fluid boluses should be guided by an assessment of fluid responsiveness (see [Chapter 11](#)).

The liberal infusion of fluids is commonplace in septic shock, but fluid accumulation has a negative impact on survival ([33,34](#)). Therefore, after the initial period of volume resuscitation, the daily infusion of fluids should be adjusted to match daily fluid losses, as this will increase the likelihood of a successful outcome.

Vasopressors

If hypotension persists after the initial volume resuscitation, infusion of a vasoconstrictor drug (vasopressor) should begin. If necessary, vasopressor infusions can begin in a peripheral vein above the antecubital fossa in the arm ([32](#)). Transition to central venous access is recommended as soon as possible, although vasopressor infusions through midline catheters have continued for 7 days without harm ([35](#)).

TABLE 17.2 The Initial Management of Septic Shock	
Therapy	Recommendations [†]
Volume Infusion	• 30 mL/kg (ideal body weight) of a crystalloid fluid, infused within 3 hrs of presentation.
Vasopressor Therapy	• Can be initiated through a peripheral vein that is proximal to the antecubital fossa.. • Begin with norepinephrine, and titrate dose to achieve an MAP $\geq$65 mm Hg. • If the required dose of norepinephrine is 0.25–0.5 μg/kg/min, add vasopressin. • If necessary, add epinephrine as a 3rd vasopressor.
Corticosteroid Therapy	• If the norepinephrine dose is >0.25 μ g/kg/min, give IV hydrocortisone, 50 mg every 6 hrs.
Antimicrobial Therapy	• Initiate treatment within 1 hr of presentation. • Obtain appropriate cultures prior to first dose of antibiotic.

[†]From Reference 32.

PREFERRED AGENTS: (Note: Vasopressor agents are described in [Chapter 14](#), and [Table 14.4](#) includes the dosing range of each agent.) *Norepinephrine is the initial vasopressor of choice in septic shock.* If the hypotension persists despite a norepinephrine dose of 0.25–0.5 μ g/kg/min (18–35 μ g/min in a 70 kg adult), then vasopressin should be added, and if the hypotension is refractory to norepinephrine and vasopressin, epinephrine is recommended as a third vasopressor

(32). The goal of vasopressor therapy is a mean arterial pressure (MAP) ≥ 65 mm Hg.

COMMENTS: The following comments about the current vasopressor recommendations deserve mention.

- . Norepinephrine has immunosuppressant effects, according to studies showing that the production of proinflammatory cytokines is suppressed during norepinephrine infusions in both laboratory animal and humans (36). The clinical significance of this effect is unclear at present, but it certainly raises a red flag about the recommendation to use norepinephrine in septic shock.
- . The recommendation for multiple vasopressors in vasopressor-refractory shock should be questioned, as there is no convincing evidence that the addition of vasopressin or a third vasopressor improves outcomes in septic shock (37,38).

Corticosteroids

Corticosteroids have two actions that are potentially beneficial in septic shock: they have anti-inflammatory activity, and they facilitate the vasoconstrictor response to catecholamines. However, innumerable studies since the 1960s have failed to produce convincing evidence that corticosteroids improve outcomes in septic shock (39). Yet steroids continue to be popular in septic shock, as shown by the following recommendation from the most recent guidelines on septic shock (32), which is based on a single meta-analysis showing that corticosteroids hastened the resolution of septic shock (by 1.5 days), but without improving the survival rate (40).

- . Corticosteroid therapy is suggested for patients with septic shock who are receiving norepinephrine or epinephrine at a dose ≥ 0.25 $\mu\text{g/kg/min}$.
- . The recommended regimen is IV hydrocortisone in a dose of 50 mg every 6 hours (200 mg daily). There is no recommendation for the duration of treatment, but most clinical studies use a treatment period of about 7 days.

The love affair with corticosteroids in septic shock continues.

Antimicrobial Therapy

Prompt initiation of antimicrobial therapy is considered a cornerstone of management in septic shock. The current recommendation is that *antimicrobial therapy should begin within 1 hour of presentation* (32). This doesn't leave much time to identify an infection, but it is important to obtain appropriate cultures (blood, urine, etc.) prior to the first dose of antimicrobial agent.

BLOOD CULTURES: At least 2 sets of blood cultures are recommended, using two separate venipuncture sites (one set can be obtained from an indwelling catheter). Two sets of blood cultures will detect about 90% of bloodstream infections, while 3 sets will detect close to 98% of bloodstream infections (41). The yield from blood cultures is influenced by the volume of blood that is cultured, and a volume of at least 20 mL is recommended for each set of blood cultures (42).

EMPIRIC COVERAGE: Empiric antibiotic coverage should have a broad spectrum of activity (e.g.,

piperacillin/tazobactam), and coverage for methicillin-resistant *Staphylococcus aureus* (MRSA) (with vancomycin or daptomycin) should be added if the patient has a prior history of MRSA infections, or MRSA infections are prevalent in the hospital. This can be discontinued if all cultures are negative for MRSA, including a nasal swab. If infection with a multidrug-resistant organism is suspected, empiric therapy should include two antimicrobial agents with gram-negative coverage (e.g., cefepime plus piperacillin/tazobactam). Antibiotics should be discontinued if all cultures are negative and there is no other evidence of infection (32).

Goals

The immediate goals of management are similar to the ones described for clinical shock in [Chapter 14](#) (see [Table 14.5](#)). These goals include improvements in tissue perfusion (e.g., PCO_2 gap <6 mm Hg) and in end-organ function (e.g., increase in urine output to >0.5 mL/kg/hr), and normalization of the plasma lactate level (<2 mmol/L).

TOXIC SHOCK SYNDROME

Toxic shock syndrome (TSS) is a specialized form of septic shock that is caused by toxin-producing strains of *Staphylococcus aureus* and *Streptococcus pyogenes* (group A streptococci), and is characterized by rapidly progressive multiorgan failure. There are differences in the TSS associated with staphylococci and streptococci, and these are shown in [Table 17.3](#).

Pathogenesis

The toxins released by the responsible microbes are capable of activating T-cell lymphocytes on a massive scale, creating a “cytokine storm” that is capable of widespread inflammatory injury. The culmination of this process is multiorgan failure and disseminated intravascular coagulopathy (DIC).

The hemodynamic changes in TSS are similar to those in septic shock, with hypotension as a result of systemic vasodilation. Disruption of the endothelium is pronounced in TSS, and the resulting fluid extravasation can lead to profound hypovolemia.

Staphylococcal TSS

Staphylococcal TSS can arise from colonization or invasive infection, and the responsible organism can be methicillin-sensitive or methicillin-resistant organisms (although the former is more frequently involved). About 40% of cases occur during menstruation (43), presumably as a result of vaginal colonization with toxin-producing staphylococci, combined with disruption of the vaginal mucosa from tampon use. Non-menstrual cases can arise from soft tissue infections such as surgical wound infections and subcutaneous abscesses. There is no identifiable infection in about 35% of cases (44).

Streptococcal TSS

Streptococcal TSS is a complication of invasive group A streptococcal infections, most notably necrotizing fasciitis, myositis, and postpartum sepsis (puerperal fever). There is no identifiable infection in almost half of the cases (45).

TABLE 17.3 **Comparative Features of the Toxic Shock Syndromes**

Feature	Staphylococcal TSS	Streptococcal TSS
Notable Sources	Menstruation (tampons)Surgical wounds No source (35%)	Necrotizing fasciitis Postpartum sepsis No source (45%)
Positive Blood Cultures	<5% of cases	60% of cases
Antibiotic Therapy	MSSA: cefazolin + clindamycin MRSA: vancomycin + clindamycin	High-dose penicillin + clindamycin
Mortality Rate	Menstrual TSS: <2% Nonmenstrual TSS: 6%	35%

From References 43,46, and 47.

Clinical Presentation

TSS typically begins with an influenza-like illness (i.e., fever, myalgias, lethargy), often associated with diarrhea and a diffuse erythematous rash (resembles a sunburn) that involves the palms and soles. (Desquamation of the skin appears after 2 to 3 weeks.) This progresses within 48 hours to hypotension and involvement of multiple organs that can include the central nervous system (altered mentation) lungs (acute respiratory distress syndrome), kidneys (acute kidney injury), liver (elevated transaminases and bilirubin), skeletal muscle (rhabdomyolysis) and the hematopoietic system (DIC). Cyanosis of the distal extremities can appear, and fluid extravasation through leaky capillaries can produce nonpitting edema.

Soft tissue infections involving group A streptococci can produce severe pain (usually in the limbs, abdomen, or pelvis) without associated findings on physical examination. Pain that is out of proportion to the physical exam should raise suspicion for a deep-seated streptococcal infection.

Diagnosis

Any case of septic shock associated with an erythematous rash and rapid-onset multiorgan failure should raise suspicion for TSS, especially if there is an association with menstruation or pregnancy. However, the rash in TSS can be subtle and fleeting (46), and TSS can be mistaken for a severe case of septic shock with multiorgan failure.

Blood cultures are positive in only 5% of cases of staphylococcal TSS (43), so the diagnosis relies heavily on an association with menstruation, and any evidence of a staphylococcal infection (e.g., subcutaneous abscesses). Blood cultures have a higher yield in streptococcal TSS, and are positive in about 60% of cases (43). In cases where a streptococcal soft tissue infection is suspected, magnetic resonance imaging (MRI) can help to uncover a covert fasciitis, and bedside ultrasound is a suitable alternative for patients that are too unstable for transport to the MRI suite.

Management

The hemodynamic management of TSS is similar to that described for septic shock, with aggressive volume resuscitation followed by vasopressors if needed. Most patients with TSS also require intubation and mechanical ventilation (47).

Antimicrobial Therapy

The rapidly progressive nature of TSS emphasizes the need for rapid initiation of antibiotic therapy. Empiric antibiotic coverage should include a broad-spectrum agent (e.g., piperacillin/tazobactam or meropenem) and enhanced gram-positive coverage with vancomycin, plus clindamycin (46). The latter agent is used in all cases of TSS because of its ability to inhibit toxin production (48). Targeted antibiotic therapy for staphylococcal and streptococcal TSS is shown in Table 17.3.

Source Control

Removing the primary source is an important component of the management of TSS. For staphylococcal TSS, this includes removing tampons and intrauterine devices, eradicating nasal colonization, and draining any identified abscesses. For streptococcal TSS, debridement of necrotizing soft tissue infections is imperative for a successful outcome.

Outcomes

Most patients survive staphylococcal TSS: the mortality rate is <2% in cases associated with menstruation, and 6% in nonmenstrual cases (46). Streptococcal TSS has far fewer survivors, with a mortality rate of 35% (47).

ANAPHYLAXIS

Anaphylaxis is an acute multiorgan dysfunction syndrome produced by the immunogenic release of inflammatory mediators from basophils and mast cells. The characteristic feature is an exaggerated immunoglobulin E (IgE) response to an external antigen; i.e., a *hypersensitivity reaction*. These reactions typically involve the skin, lungs, gastrointestinal tract, and cardiovascular system, and common triggers include drugs (especially antimicrobial agents), food, and insect bites. Contrary to popular perception, deaths from anaphylaxis are rare, with a mortality rate of 0.25%–0.33% (49).

Clinical Features

Anaphylactic reactions typically appear within minutes of exposure to an external trigger, and they can recur within 72 hours of the initial episode (biphasic anaphylaxis) (49). A characteristic feature of these reactions is edema and swelling in the involved organ, which is the result of an increase in vascular permeability. The clinical manifestations of anaphylactic reactions are shown in Table 17.4 (50). Common manifestations include urticaria, subcutaneous angioedema (typically involving the face), flushing, angioedema of the upper airway (e.g., laryngeal edema) and bronchospasm. The most feared manifestation is hypotension (i.e., anaphylactic shock), and the most overlooked manifestation is an acute coronary syndrome (see next).

TABLE 17.4 Signs and Symptoms of Anaphylaxis

Manifestation	Frequency of Occurrence
Urticaria and angioedema	60–90%
Upper airway angioedema	50–60%
Flushing	45–55%
Dyspnea, wheezing	45–50%
Hypotension, syncope	30–35%
Abdominal pain, diarrhea	25–30%
Acute Coronary Syndrome	4–5%

From Reference 50.

Allergic Coronary Insufficiency

The association of anaphylaxis and acute coronary syndromes (the Kounis Syndrome) is attributed to coronary vasospasm (in 73% of cases), coronary artery plaque rupture (in 22% of cases) and coronary stent thrombosis (in 5% of cases) (51). The presumed mechanisms include the release of vasoactive substances from mast cell degranulation, and platelet activation. Coronary vasospasm is the most common underlying mechanism, and occurs predominantly in younger patients with no history of coronary artery disease. Cardiac symptoms usually appear within one hour of the anaphylactic reaction (e.g., urticaria, angioedema, bronchospasm), and can include fleeting chest pains (from coronary spasm) or ST-elevation myocardial infarction (from plaque rupture or stent thrombosis).

Treatment

The treatment of anaphylaxis includes one drug that halts the reaction (epinephrine), and drugs that alleviate symptoms (antihistamines, bronchodilators).

Epinephrine

Epinephrine blocks the release of inflammatory mediators from sensitized basophils and mast cells, and is *the only drug capable of stopping an anaphylactic reaction* (50). It is available in a confusing array of aqueous solutions, which are shown in Table 17.5. The optimal epinephrine dose for anaphylaxis is unknown, but the usual dose in adults is 0.3–0.5 mg (0.3–0.5 mL of 1:1000 solution) given by deep intramuscular (IM) injection in the anterolateral thigh, and repeated every 5–15 minutes if necessary (50). Drug absorption is slower with subcutaneous injection (52). Epinephrine can be nebulized for laryngeal edema using the dosing regimen in Table 17.5, but the efficacy of this regimen is unclear.

TABLE 17.5 Aqueous Epinephrine Solutions and Their Clinical Uses

Aqueous Dilution	Condition	Dosing Regimen

1:100 (10 mg/mL)	Laryngeal Edema	0.25 mL (2.5 mg) in 2 mL saline and administer by nebulizer.
1:1,000 (1 mg/mL)	Anaphylaxis	0.3–0.5 mL (mg) by deep IM injection in the thigh every 5–15 min as needed.
1:10,000 (0.1 mg/mL)	Asystole or PEA	10 mL (1 mg) IV every 3–5 min as needed.
1:100,000 (10 µg/mL)	Anaphylactic Shock	Add 2 mL of 1:1000 solution (2 mg) to 200 mL saline (10 µg/mL) and infuse at 5–15 µg/min.

From Reference 53.

GLUCAGON: The inhibitory actions of epinephrine on mast cell degranulation are mediated by β -adrenergic receptors, and ongoing therapy with β -receptor antagonists can attenuate or eliminate the response to epinephrine. When anaphylactic reactions are refractory to epinephrine in patients receiving β -blocker drugs, glucagon can be effective in restoring responsiveness (for reasons described in [Chapter 52](#)). The dose of glucagon is 1–5 mg by slow intravenous injection (over 5 min), followed by a continuous infusion at 5–15 µg/min, titrated to the desired response ([53](#)). Glucagon can trigger vomiting, and patients with depressed consciousness should be placed on their side when glucagon is administered, to limit the risk of aspiration.

Second-Line Agents

The following drugs are used for symptom relief, and do not alter the course of the illness. They are given after epinephrine is administered, and should never be used as a replacement for epinephrine.

ANTIHISTAMINES: Histamine receptor antagonists are used to relieve the pruritis and rhinorrhea associated with anaphylactic reactions (although their efficacy is unproven). The histamine H_1 blocker *diphenhydramine* (25–50 mg PO, IM, or IV) and the histamine- H_2 blocker *ranitidine* (50 mg IV or 150 mg PO) are considered equivalent in efficacy, but combination therapy produces the best results ([54](#)).

BRONCHODILATORS: Inhaled β_2 -receptor agonists like *albuterol* are used to relieve bronchospasm, and are administered by nebulizer (2.5 mL of a 0.5% solution) or by metered-dose inhaler ([54](#)).

Corticosteroids

Corticosteroids have enjoyed a longstanding popularity in the treatment of hypersensitivity reactions, and they are sometimes preferred to epinephrine for the acute management of anaphylaxis ([54](#)). However, there is no evidence that corticosteroids are effective in reversing, slowing, or preventing the recurrence of anaphylactic reactions ([49,50,54](#)). This is reflected in the following statement from a clinical practice guideline on anaphylaxis ([50](#)): “*The use of corticosteroids has no role in the acute management of anaphylaxis*”.

Anaphylactic Shock

Anaphylactic shock is characterized by the same hemodynamic abnormalities described earlier

for septic shock, but the hypotension in anaphylactic shock can be more pronounced because of the exaggerated fluid loss through leaky capillaries. Management includes intravenous epinephrine and aggressive volume resuscitation.

Epinephrine

There is no standardized dosing regimen for epinephrine in anaphylactic shock, but the dose rate of 5–15 µg/min in [Table 17.5](#) has been cited for its efficacy ([53](#)). A bolus dose of 10 µg IV can precede the drug infusion, and can be repeated periodically if needed to maintain the mean arterial pressure at ≥65 mm Hg ([54](#)).

Volume Resuscitation

Aggressive volume resuscitation is essential in anaphylactic shock because as much as 35% of the intravascular volume can be lost through leaky capillaries ([53](#)), which is enough to produce hypovolemic shock (see [Table 15.1](#)). Volume resuscitation can begin by infusing 1–2 liters of a crystalloid fluid (or 20–30 mL/kg), or 500 mL of 5% albumin, over the first 5–10 minutes ([53](#)). Thereafter, fluid management should be tailored to the clinical condition of the patient. Considering adding a colloid fluid (e.g., 5% albumin) to your volume regimen, since the severity of the capillary leak in anaphylactic shock will ensure that most of the infused crystalloid fluid will move out of the vascular compartment and add to the edema.

Refractory Hypotension

For hypotension that persists despite epinephrine and volume resuscitation, give glucagon if the patient has received a β blocker recently. Otherwise, add a second vasopressor like norepinephrine or phenylephrine (vasopressin has not been evaluated in anaphylactic shock). The prognosis in this situation is poor.

A FINAL WORD

Inflammation (not hypoxia) as the Major Cause of Death in ICUs

The following information is a spinoff of the information presented in A FINAL WORD in [Chapter 9](#), which is titled “Is cellular hypoxia a common cause of death?” The observations presented in this chapter, which are summarized in the statements below, provide evidence that inflammation, not cellular hypoxia, is the most common cause of death in the hospital.

- . Septic shock is the leading cause of in-hospital deaths in the United States ([16](#)).
- . The source of organ failure in septic shock is mitochondrial dysfunction, not inadequate tissue oxygenation (see [Figure 17.3](#)), and the source of the mitochondrial dysfunction is inflammation.

The apparent role of inflammation in fatal outcomes creates doubt about the emphasis on promoting tissue oxygenation (e.g., with O₂ inhalation and RBC transfusions) in critically ill patients. In fact, since the damaging effects of inflammation are largely due to oxidation, promoting tissue oxygenation may have an effect that is diametrically opposed to what is

intended. More on this in [Chapter 25 \(Oxygen Therapy\)](#).

References

- a. Hunter J. A Treatise on the Blood, Inflammation, and Gun-shot Wounds. London: Nicol, 1794.
1. Moore W. The Knife Man. Blood, Body Snatching, and the Birth of Modern Surgery. New York: Broadway Books, 2005.
2. Turk JL. Inflammation: John Hunter's A Treatise on the Blood, Inflammation, and Gun-shot Wounds. *Int J Exp Path* 1994; 75:385–395.

Inflammatory Injury

3. Pinsky MR, Matuschak GM. Multiple systems organ failure: failure of host defense mechanisms. *Crit Care Clin* 1989; 5:199–220.
4. Pinsky MR, Vincent J-L, Deviere J, et al. Serum cytokine levels in human septic shock: Relation to multiple-system organ failure and mortality. *Chest* 1993; 103:565–575.
5. Hurst JK, Barrette WC. Leukocyte oxygen activation and microbicidal oxidative toxins. *Crit Rev Biochem Molec Biol* 1989; 24:271–328.
6. Winkelstein JA, Marino MC, Johnston RB, et al. Chronic granulomatous disease. Report on a national registry of 368 patients. *Medicine (Baltimore)* 2000; 79:155–169.
7. Mittal M, Siddiqui MR, Tran K, et al. Reactive oxygen species in inflammation and tissue injury. *Antiox Redox Signal* 2014; 20:1126–1167.
8. Chuang CY, Degendorfer G, Davies MJ. Oxidation and modification of extracellular matrix and its role in disease. *Free Rad Res* 2014; 48:970–989.
9. Marino PL. Does oxygen participate in the inflammatory response? In: *Oxygen: Creating a New Paradigm*. Philadelphia: Wolters Kluwer, 2022;123–137.
10. Wang Z-Q, Porreca F, Cuzzocrea S, et al. A newly identified role for superoxide in inflammatory pain. *J Pharmacol Exp Therap* 2004; 309:869–878.
11. Montezano AC, Touyz RM. Reactive oxygen species and endothelial function – role of nitric oxide synthase uncoupling and Nox family nicotinamide adenine dinucleotide phosphate oxidases. *Basic Clin Pharmacol Toxicol* 2011; 110:87–94.
12. Huet O, Obata R, Aubron C, et al. Plasma-induced endothelial oxidative stress is related to the severity of septic shock. *Crit Care Med* 2007; 35:821–826.
13. Radi R. Oxygen radicals, nitric oxide, and peroxynitrite: Redox pathways in molecular medicine. *Proc Nat Acad Sci* 2018; 115:5839–5848.
14. Angus DC, Linde-Zwirble WT, Lidicker J, et al. Epidemiology of severe sepsis in the United States: Analysis of incidence, outcome, and associated costs of care. *Crit Care Med* 2001; 29:1303–1310.
15. Vincent J-L, de Mendonca A, Cantraine F, et al. Use of the SOFA score to assess the incidence of organ dysfunction/failure in intensive care units: Results of a multicenter, prospective study. *Crit Care Med* 1998; 26:1793–1800.

Septic Shock

16. Liu V, Escobar GJ, Greene JD, et al. Hospital deaths in patients with sepsis from two independent cohorts. *JAMA* 2014; 312:90–92.
17. Rudd KE, Johnson SC, Agesa KM, et al. Global, regional, and national sepsis incidence and mortality, 1990–2017: analysis for the Global Burden of Disease Study. *Lancet* 2020; 395:200–211.
18. Paoli CJ, Reynolds MA, Sinha M, et al. Epidemiology and costs of sepsis in the United States – an analysis based on timing of diagnosis and severity level. *Crit Care Med* 2018; 46:1889–1897.
19. Zahar J-R, Timsit J-F, Garrouste-Orgeas M, et al. Outcomes in severe sepsis and patients with septic shock: pathogen species and infection sites are not associated with mortality. *Crit Care Med* 2011; 39:1886–1895.
20. Singer M, Deutschman CS, Seymour CW, et al. The third international consensus definitions for sepsis and septic shock (Sepsis-3). *JAMA* 2016; 315:801–810.
21. American College of Chest Physicians/Society of Critical Care Medicine Consensus Conference Committee. Definitions of sepsis and organ failure and guidelines for the use of innovative therapies in sepsis. *Chest* 1992; 101:1644–1655.
22. Pittet D, Rangel Frausto S, Li N, et al. Systemic inflammatory response syndrome, sepsis, severe sepsis, and septic shock: incidence, morbidities and outcomes in surgical ICU patients. *Intensive Care Med* 1995; 21:302–309.
23. Rangel Frausto MS, Pittet D, Costigan M, et al. Natural history of the systemic inflammatory response syndrome (SIRS). *JAMA* 1995; 273:117–123.
24. Abraham E, Singer M. Mechanisms of sepsis-induced organ dysfunction. *Crit Care Med* 2007; 35:2409–2416.
25. Snell RJ, Parillo JE. Cardiovascular dysfunction in septic shock. *Chest* 1991; 99:1000–1009.
26. Fink MP. Cytopathic hypoxia. Mitochondrial dysfunction as mechanism contributing to organ dysfunction in sepsis. *Crit Care Clin* 2001; 17:219–237.
27. Ruggieri AJ, Levy RJ, Deutschman CS. Mitochondrial dysfunction and resuscitation in sepsis. *Crit Care Clin* 2010; 26:567–575.

28. Muravchick S, Levy RJ. Clinical implications of mitochondrial dysfunction. *Anesthesiology* 2006; 105: 819–837.
29. Thomas GW, Mains CW, Slone D, et al. Potential dysregulation of the pyruvate dehydrogenase complex by bacterial toxins and insulin. *J Trauma* 2009; 67:628–633.
30. Sair M, Etherington PJ, Winlove CP, Evans TW. Tissue oxygenation and perfusion in patients with systemic sepsis. *Crit Care Med* 2001; 29:1343–1349.
31. Vlessis AA, Goldman RK, Trunkey DD. New concepts in the pathophysiology of oxygen metabolism during sepsis. *Br J Surg* 1995; 82:870–876.
32. Evans L, Rhodes A, Alhazzan W, et al. Surviving sepsis campaign: international guidelines for management of sepsis and septic shock, 2021. *Intensive Care Med* 2021; 47:1181–1247.
33. Boyd JH, Forbes J, Nakada T-a, et al. Fluid resuscitation in septic shock: a positive fluid balance and elevated central venous pressure are associated with increased mortality. *Crit Care Med* 2011; 39:259–265.
34. Sakr Y, Robatto PN, Kotfis K, et al, on behalf of the Intensive Care Over Nations Investigators. Higher fluid balance increases the risk of death from sepsis: results from a large international audit. *Crit Care Med* 2017; 45:386–394.
35. Prasanna N, Yamane D, Haridas N, et al. Safety and efficacy of vasopressor administration through midline catheters. *J Crit Care* 2021; 61:1–4.
36. Stolk RF, van der Pasch E, Naumann F, et al. Norepinephrine dysregulates the immune response and compromises host defense during sepsis. *Am J Respir Crit Care Med* 2020; 202:830–842.
37. Russell JA, Walley KR, Singer J, et al; for the VASST Investigators. Vasopressin versus norepinephrine infusion in patients with septic shock. *N Engl J Med* 2008; 358:877–887.
38. Oriyanka P, Chang C-CH, Chawla LS, et al. Vasopressor-resistant hypotension, combination vasopressor therapy, and shock phenotypes in critically ill adults with vasodilatory shock. *Shock* 2022; 58:260–268.
39. Venkatesh B, Finfer S, Cohen J, et al; for the ADRENAL Trial Investigators and the Australian-New Zealand Intensive Care Society Clinical Trials Group. Adjunctive glucocorticoid therapy in patients with septic shock. *N Engl J Med* 2018; 378:797–808.
40. Rygård SL, Butler E, Granholm A, et al. Low-dose corticosteroids for adult patients with septic shock: a systematic review with meta-analysis and trial sequential analysis. *Intensive Care Med* 2018; 44:1003–1016.
41. Lee A, Mirrett S, Reller B, Weinstein MP. Detection of bloodstream infections in adults: how many blood cultures are needed? *J Clin Microbiol* 2007; 45:3546–3548.
42. Cockerill FR III, Wilson JW, Vetter EA, et al. Optimal testing parameters for blood cultures. *Clin Infect Dis* 2004; 38:1724–1730.

Toxic Shock Syndrome

43. Lappin E, Ferguson AJ. Gram-positive toxic shock syndromes. *Lancet Infect Dis* 2009; 9:281–290.
44. Devries AS, Leshner L, Schlievert PM, et al. Staphylococcal toxic shock syndrome 2000–2006: epidemiology, clinical features, and molecular characteristics. *PloS One* 2011; 6:e22997.
45. Stevens DL, Tanner MH, Winship J, et al. Severe group A streptococcal infections associated with a toxic shock-like syndrome and scarlet fever toxin A. *N Engl J Med* 1989; 32:1–7.
46. Chu, VH. Staphylococcal toxic shock syndrome. [UpToDate.com](https://www.uptodate.com/contents/staphylococcal-toxic-shock-syndrome). Accessed September 15, 2023.
47. Schmitz M, Roux X, Huttner B, Pugin J. Streptococcal toxic shock syndrome in the intensive care unit. *Ann Intensive Care* 2018; 8:88.
48. Russell NE, Pachorek RE. Clindamycin in the treatment of streptococcal and staphylococcal toxic shock syndromes. *Ann Pharmacother* 2000; 34:936–939.

Anaphylaxis

49. Shaker MS, Wallace DV, Golden DBK, et al. Anaphylaxis—a 2020 practice parameter update, systematic review, and Grading of Recommendations, Assessment, Development and Evaluation (GRADE) analysis. *J Allergy Clin Immunol* 2020; 145:1082–1123.
50. Lieberman P, Nicklas RA, Randolph, et al. Anaphylaxis—a practice parameter update 2015. *Ann Allergy Asthma Immunol* 2015; 115:341–384.
51. Alblaihed L, Huis in't Veld MA. Allergic coronary syndrome—Kounis syndrome. *Emerg Med Clin N Am* 2022; 40:69–78.
52. Simons FER, Gu X, Simons KJ. Epinephrine absorption in adults: intramuscular versus subcutaneous injection. *J Allergy Clin Immunol* 2001; 108(5):871–3.
53. Lieberman P, Nicklas RA, Oppenheimer J, et al. The diagnosis and management of anaphylaxis practice parameter: 2010 update. *J Allergy Clin Immunol* 2010; 126:480.e1–480.e42.
54. McHugh K, Repanshek Z. Anaphylaxis. Emergency department treatment. *Emerg Med Clin N Amer* 2022; 40:19–32.

CARDIAC DISORDERS

Nothing is so firmly believed as that which is least known.

Francis Jeffrey
(1773–1850)